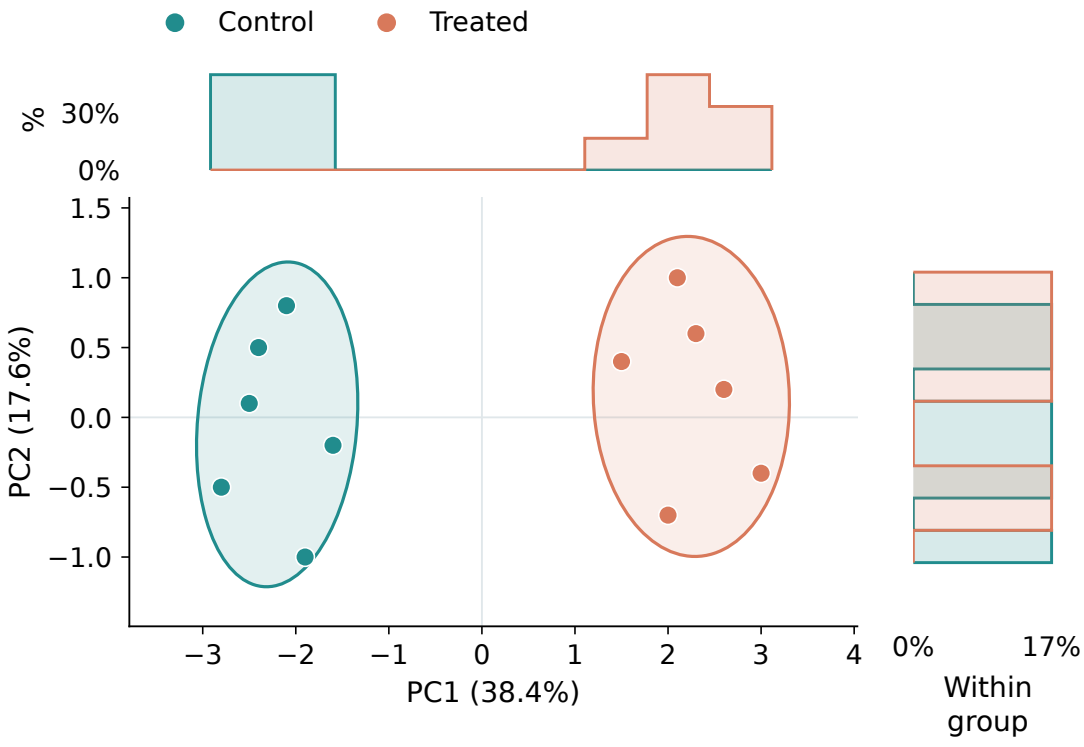
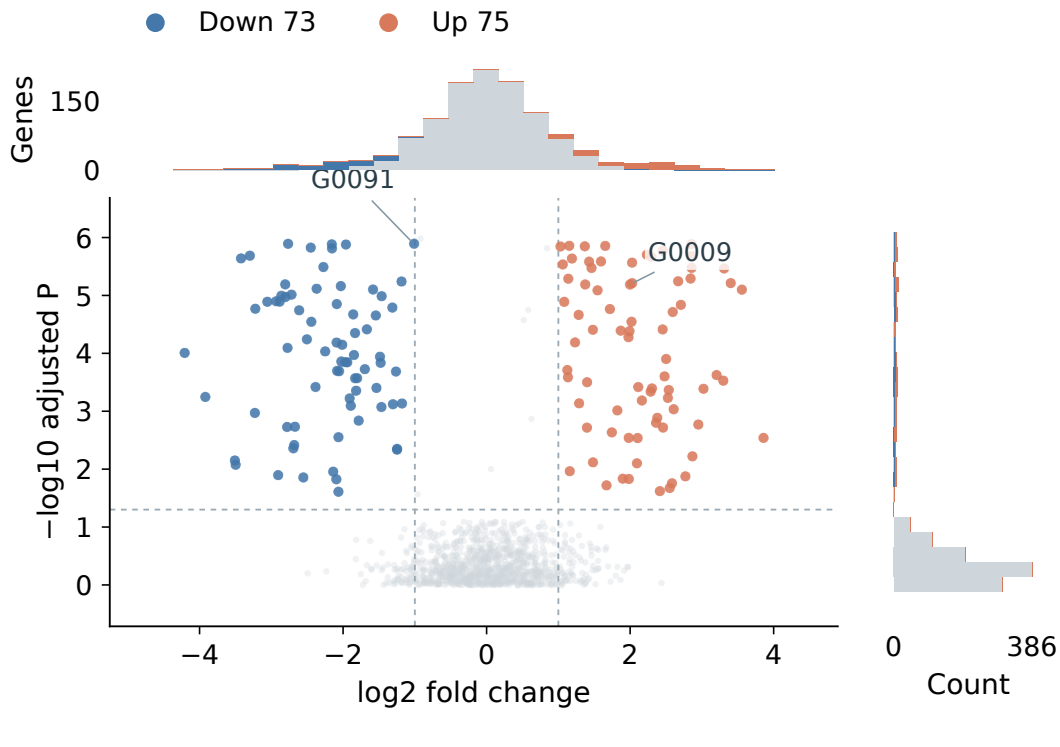
Treated versus Control / illustrative
12 synthetic donors / all results are fictional

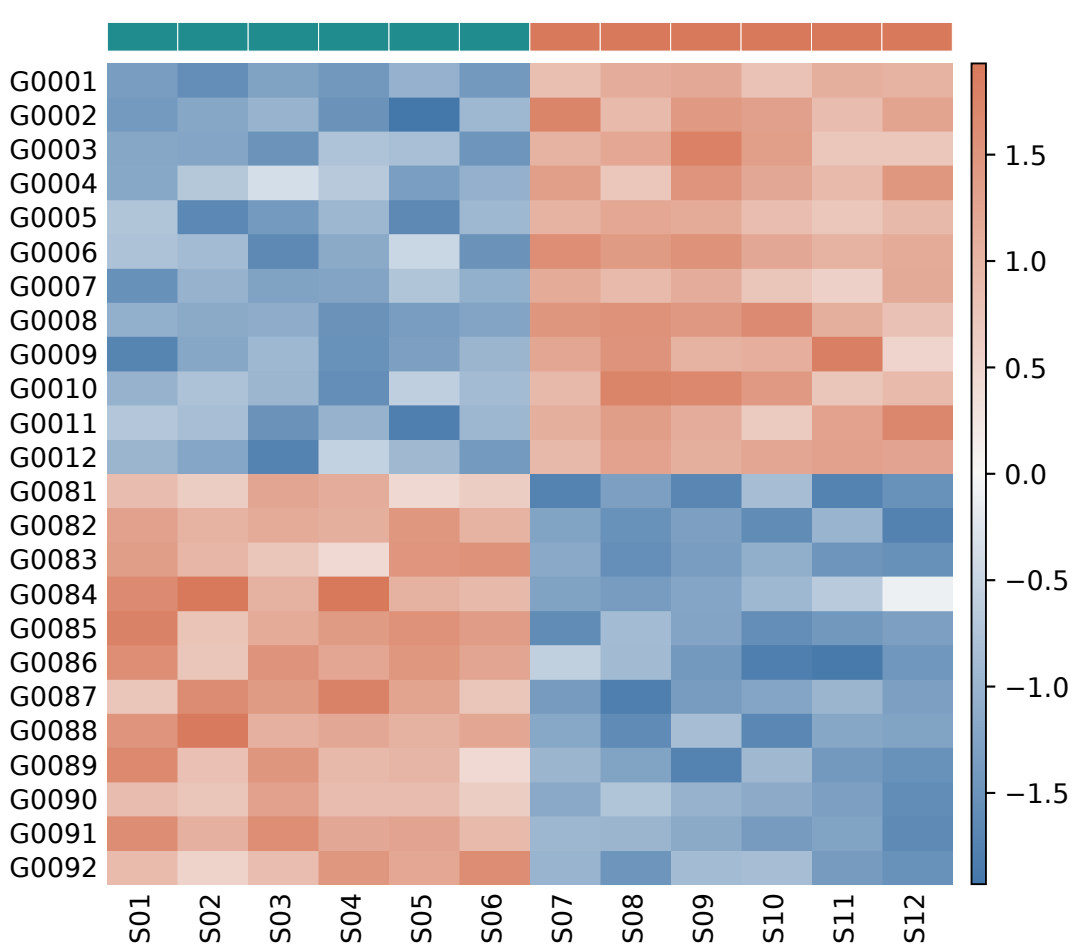
A Sample architecture



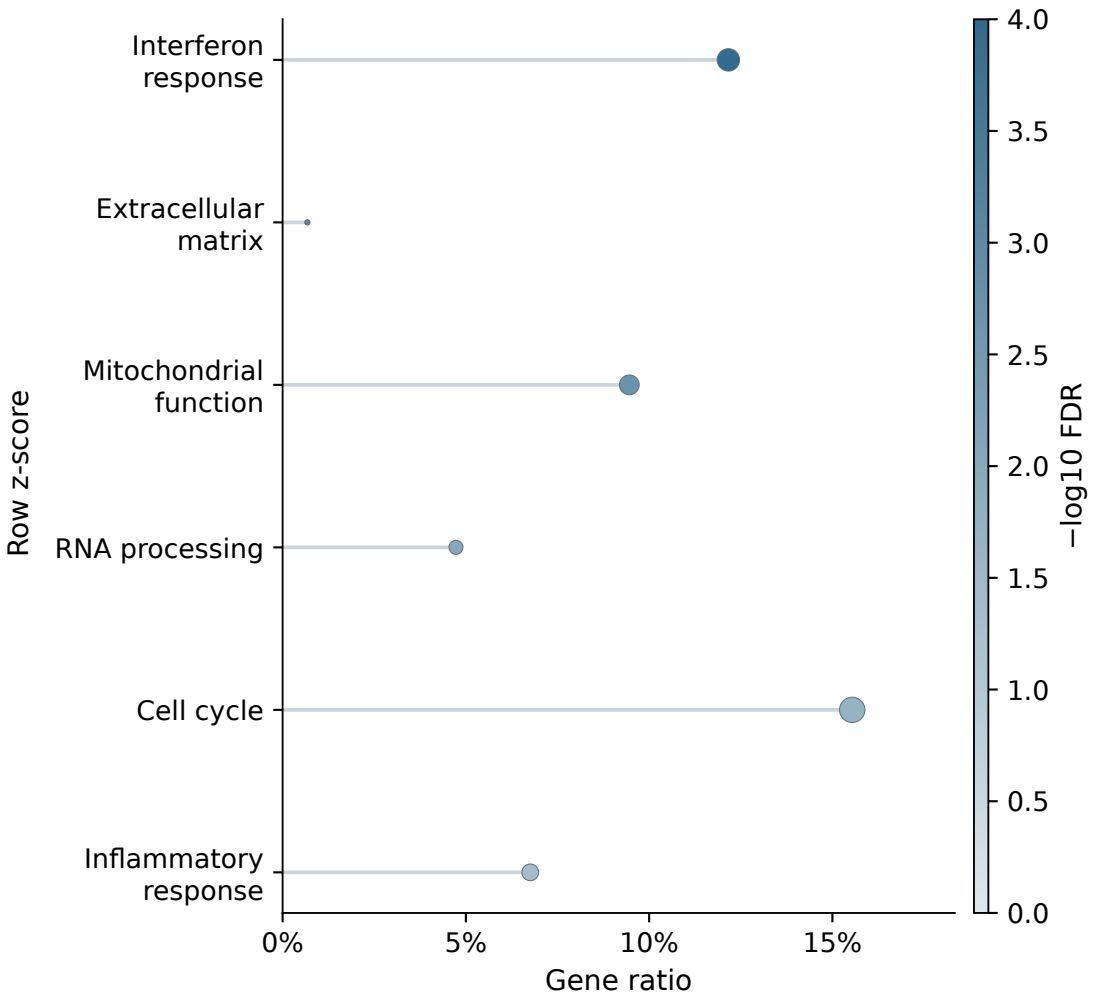
B Differential response



C Prespecified expression program

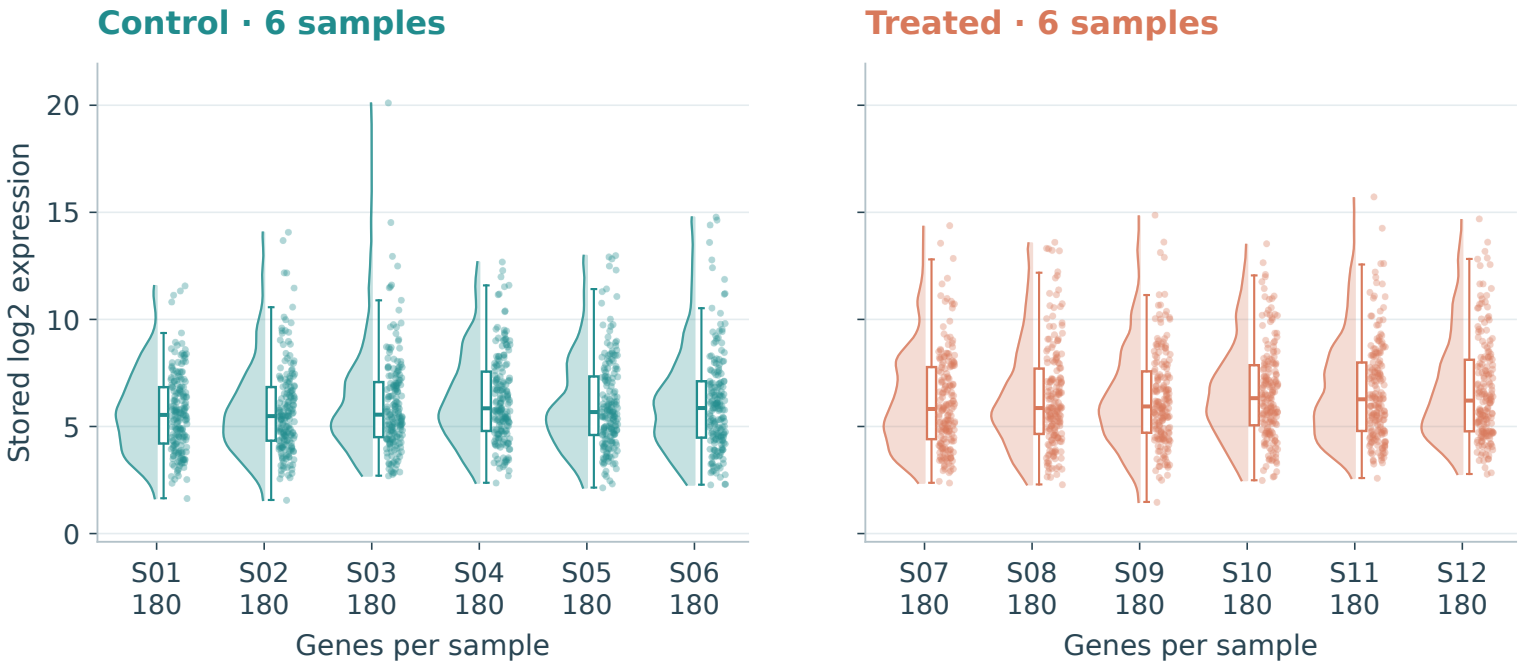


D Functional context

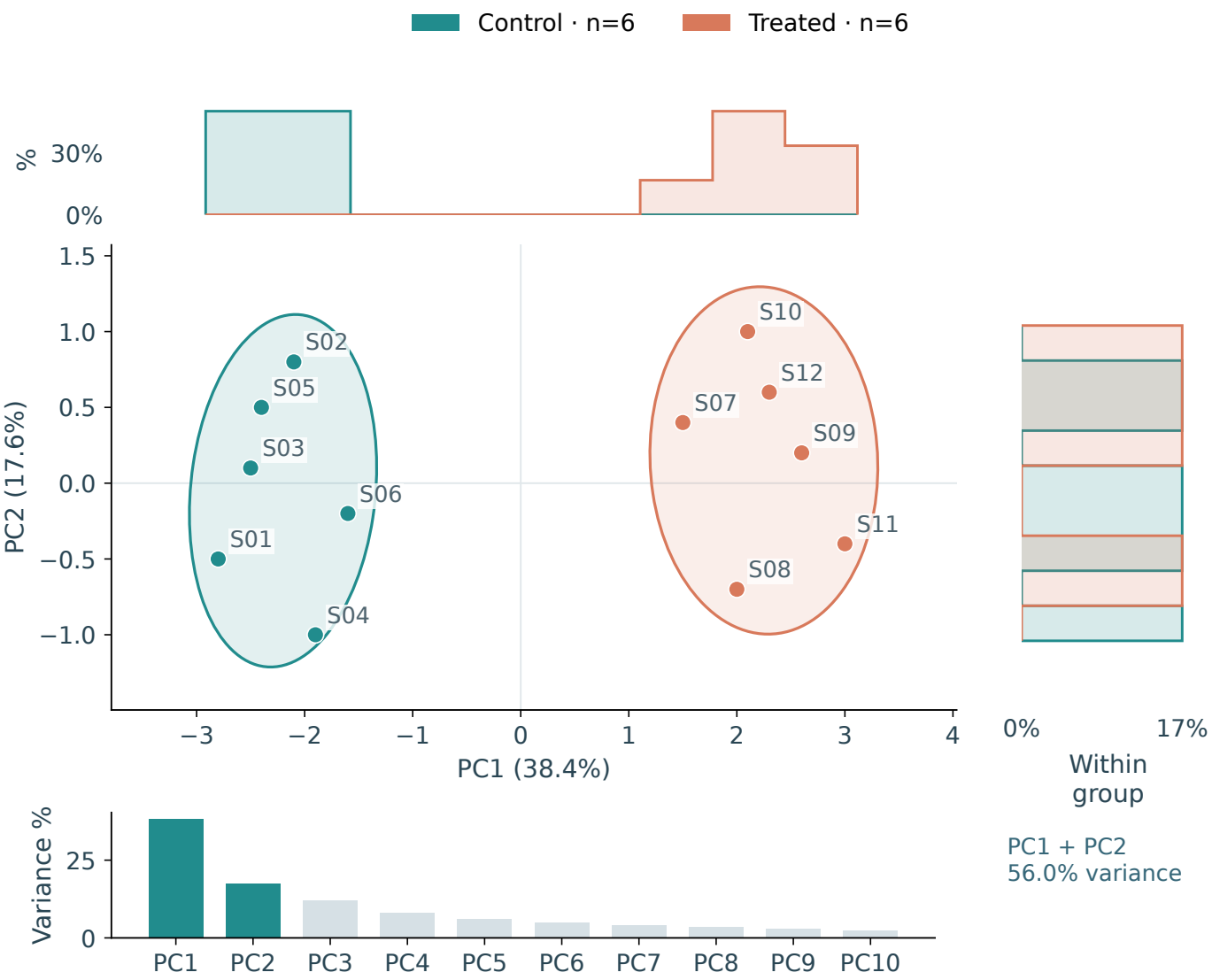


Expression distributions

12 samples · complete supplied observations · shared metric axes



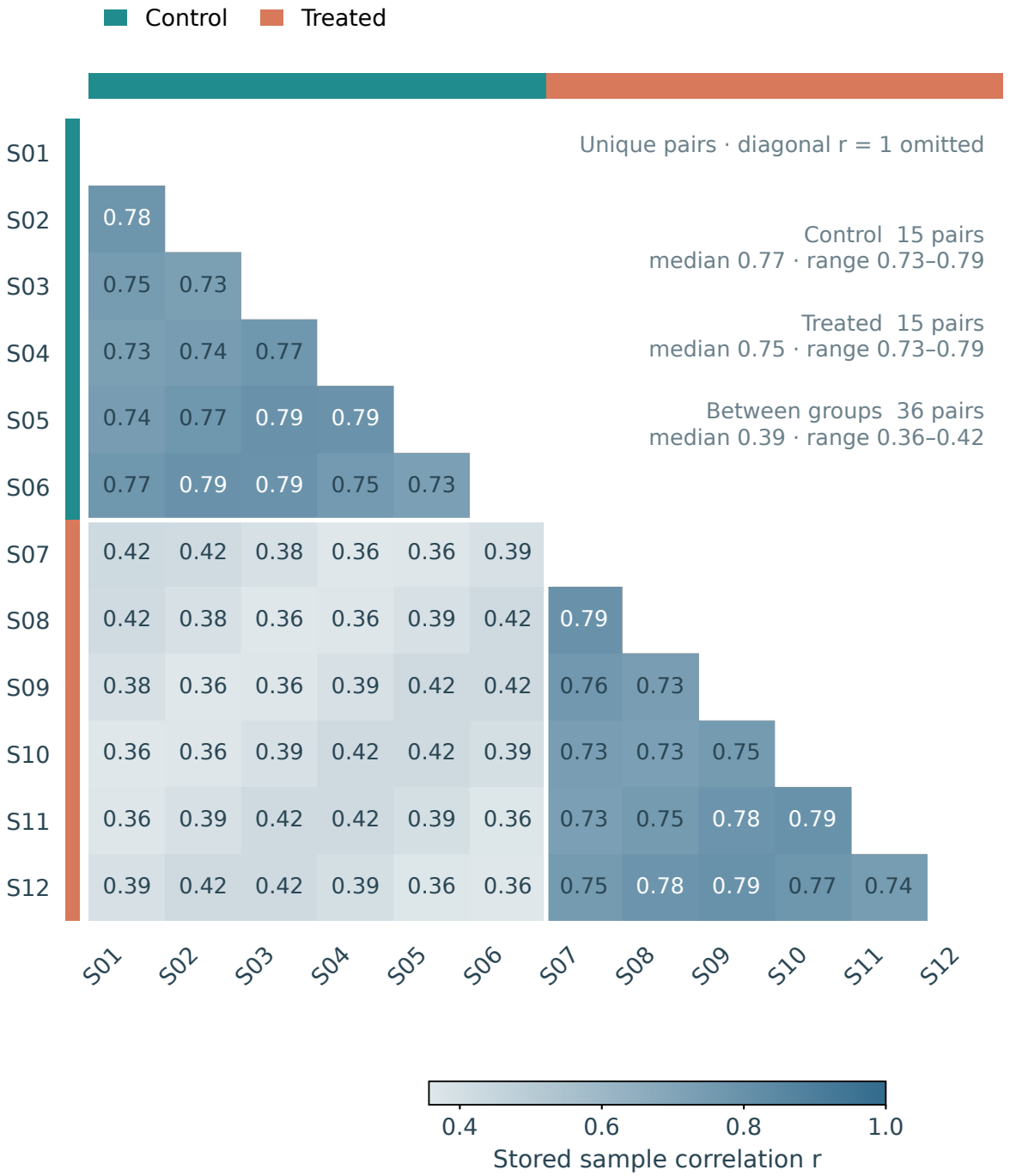
Sample structure



Supplied 95% reference dispersion contours; illustrative, not mean confidence intervals or a separation test.

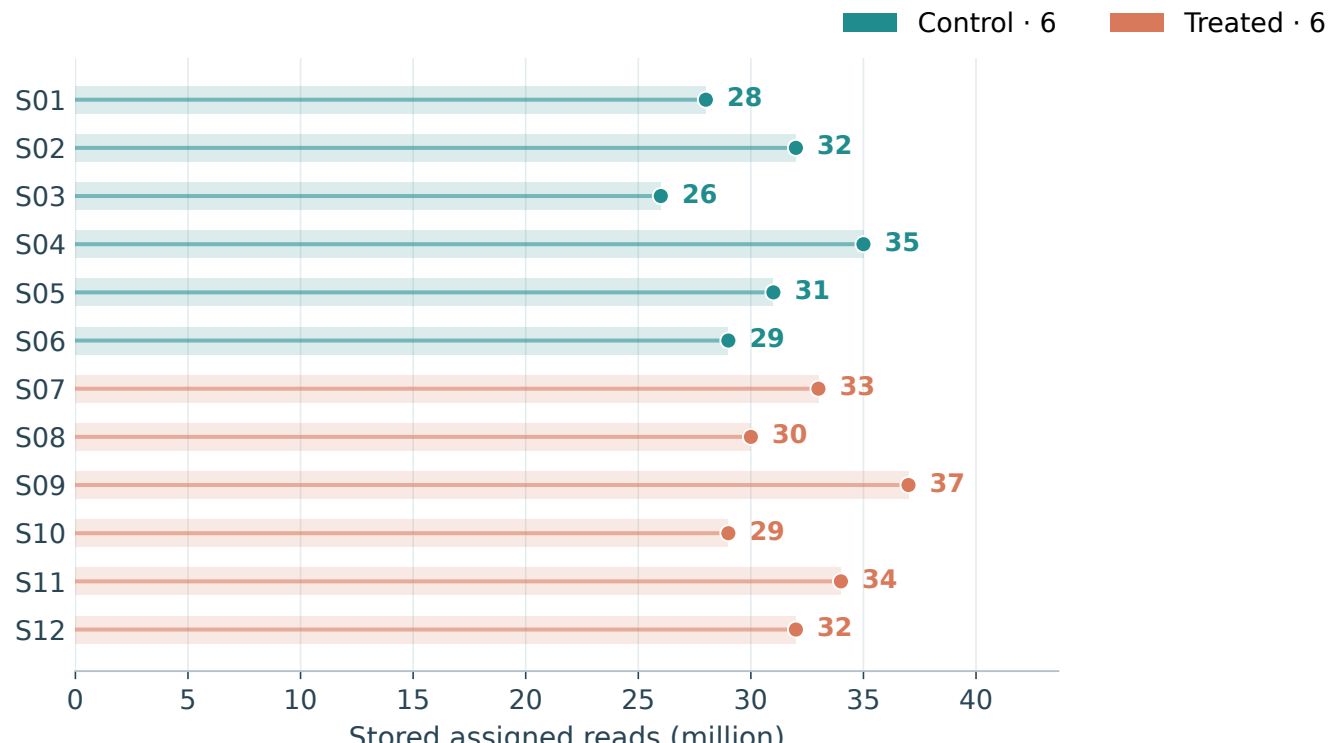
Sample correlation

Unique sample pairs



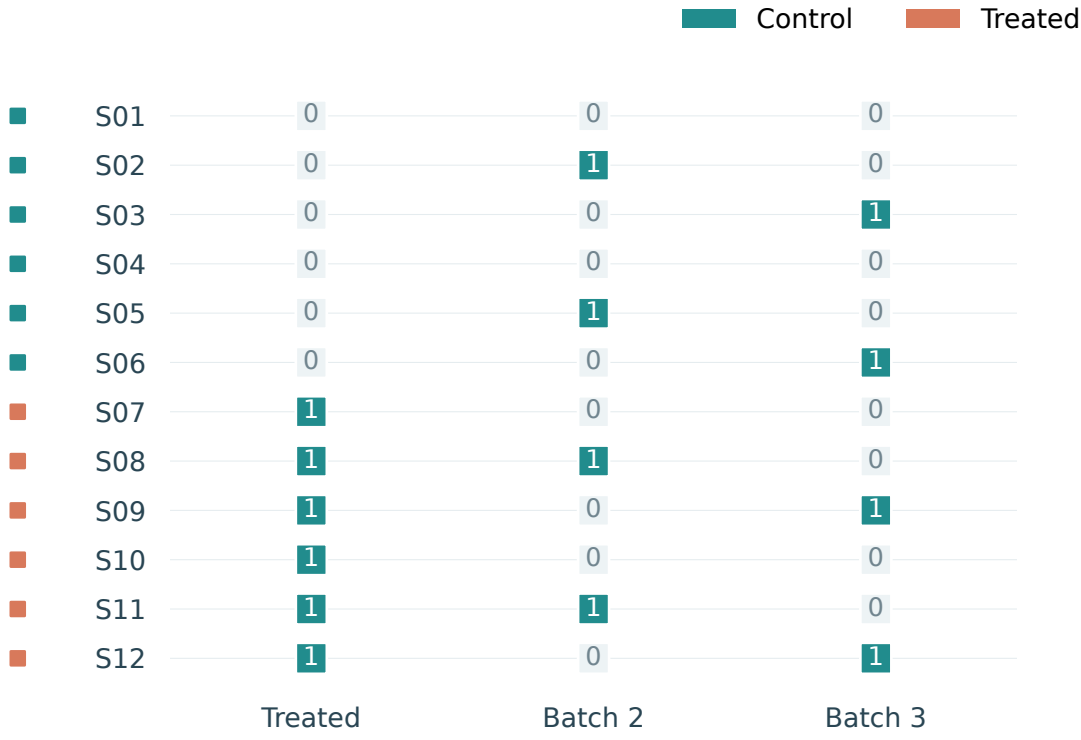
Library coverage

12 supplied units · all counts retained



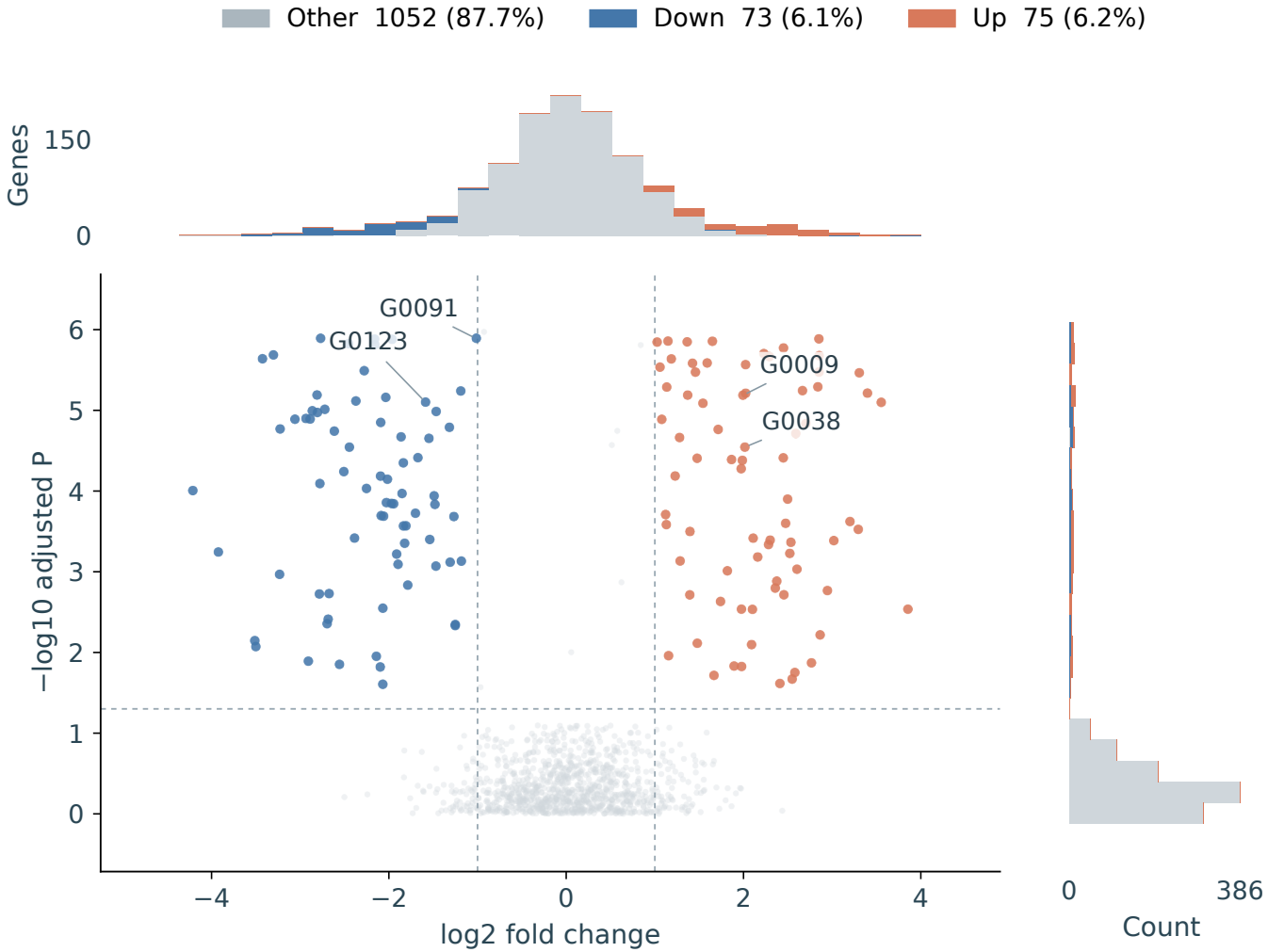
Study design and batch coding

12 simulated donors · six per condition



Reference: Control and Batch 1; 0 = absent, 1 = present

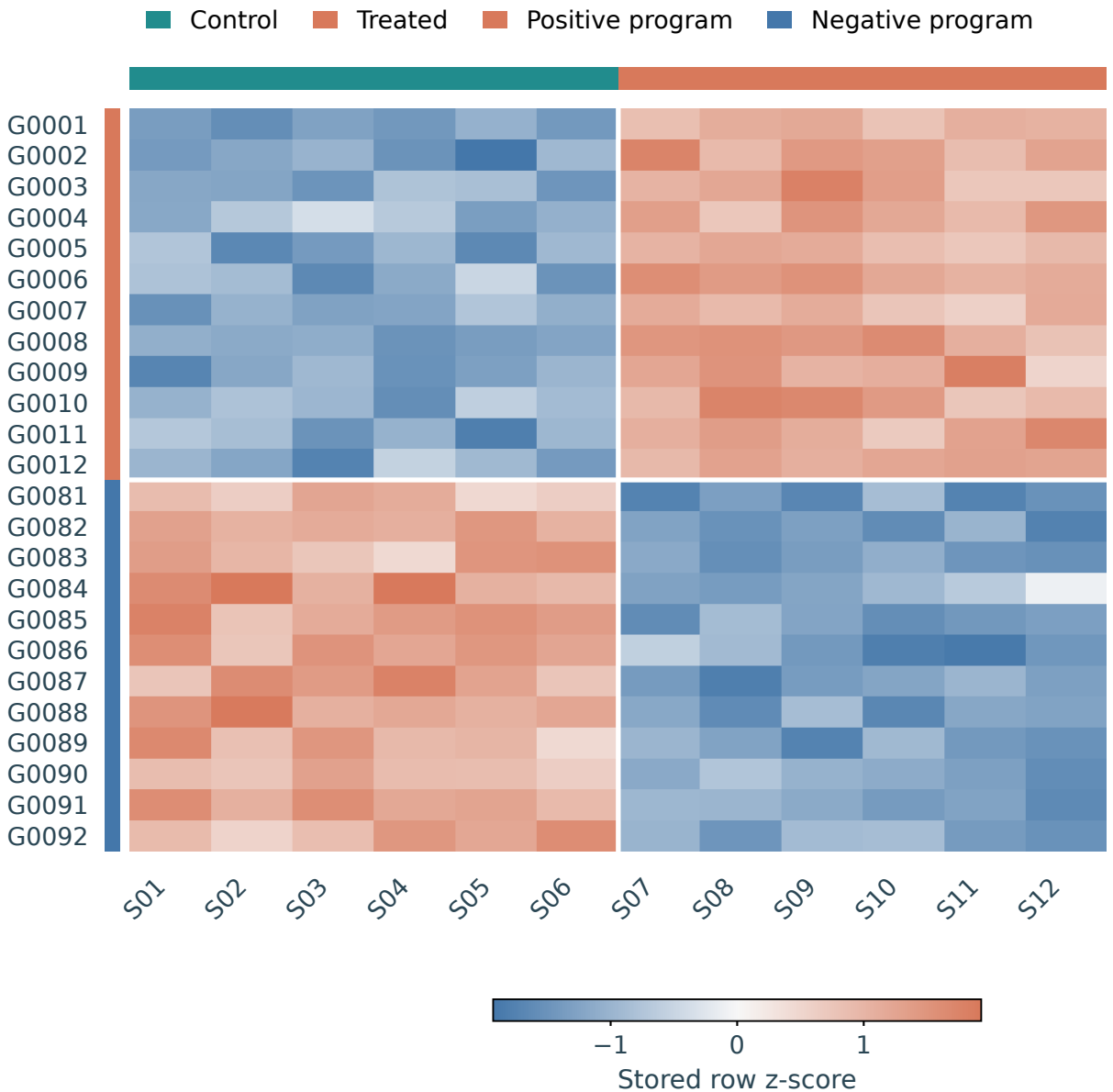
Differential expression



All 1,200 tested genes · stacked counts · FDR ≤ 0.05 and |log2FC| ≥ 1

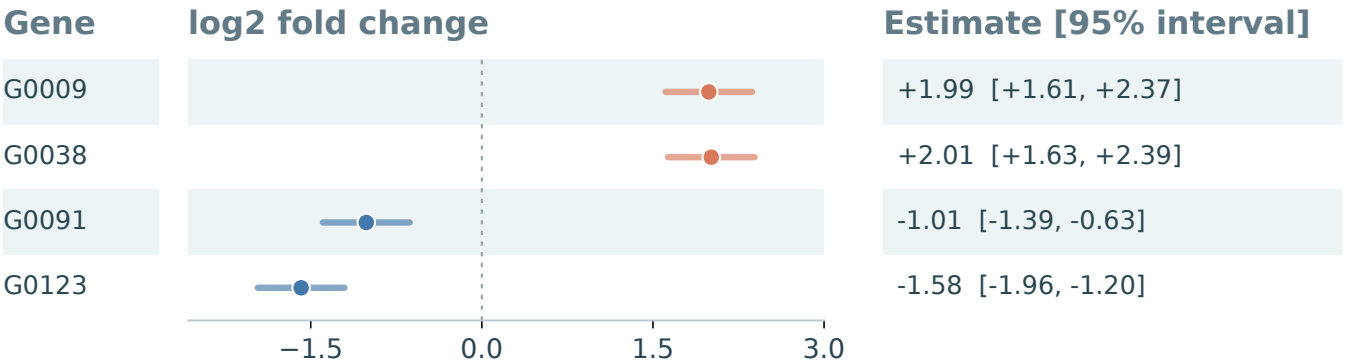
Selected expression programs

24 selected features · 12 samples

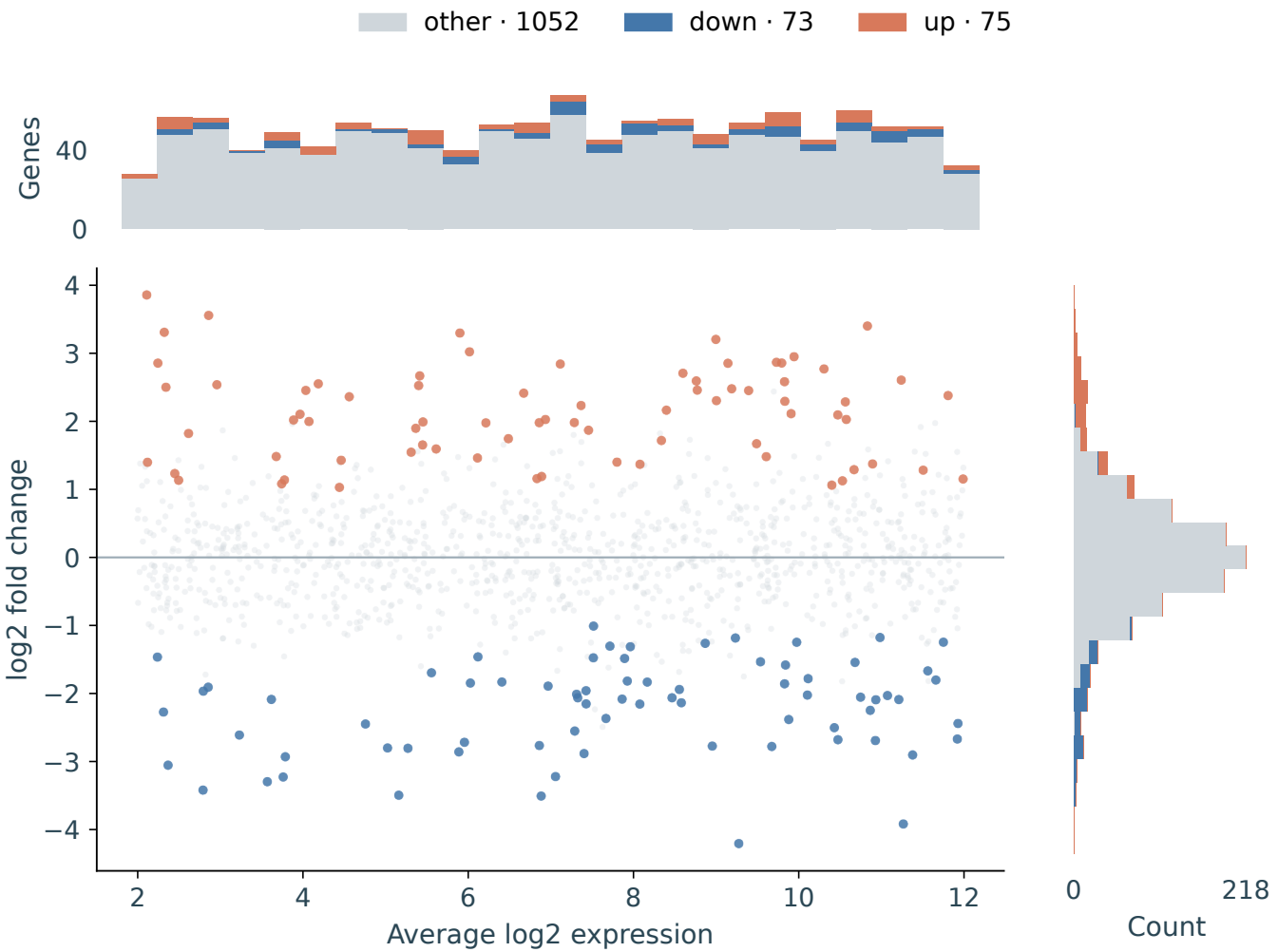


Selected gene effects

Treated versus Control / donor-level

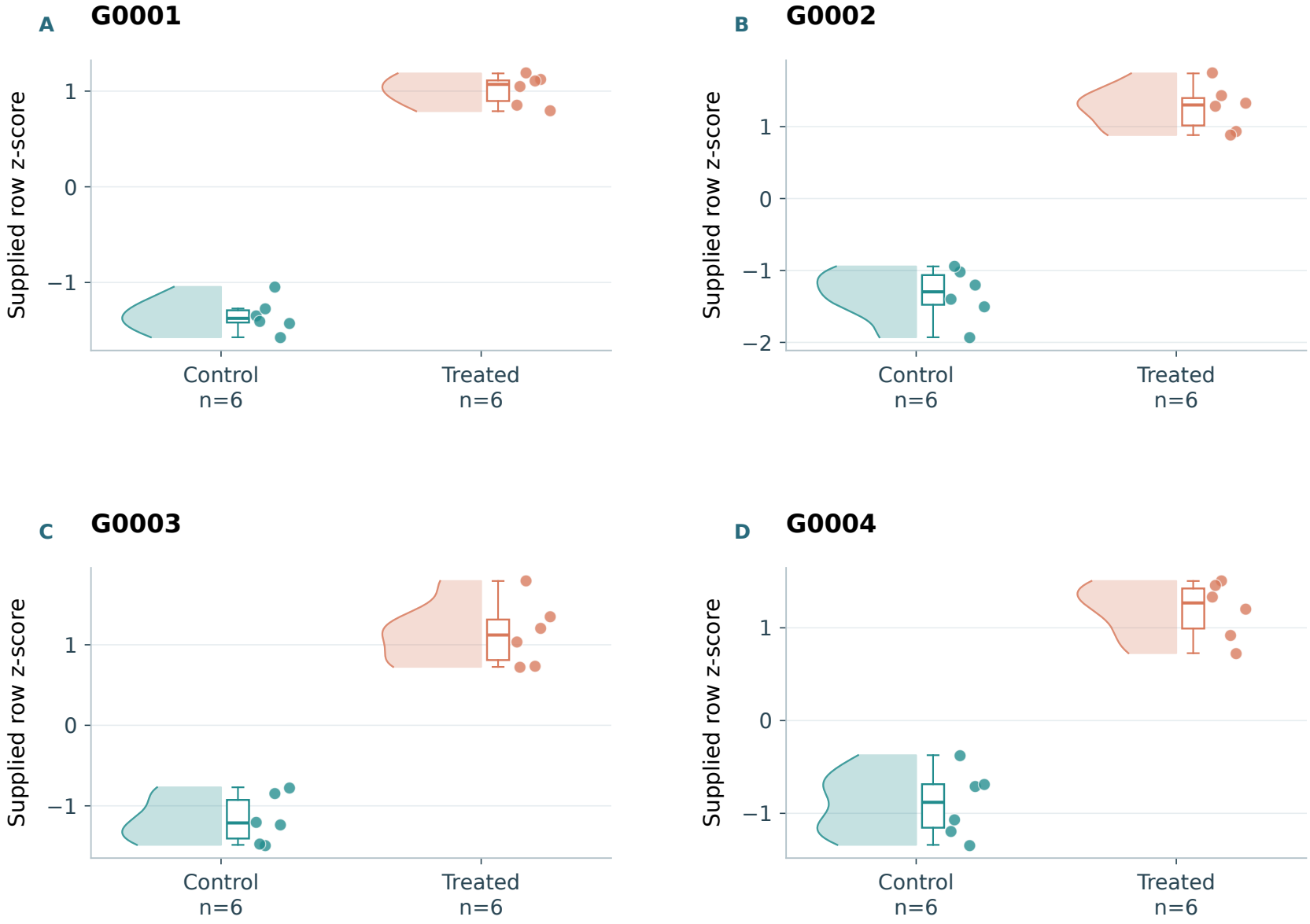


Expression level and effect size



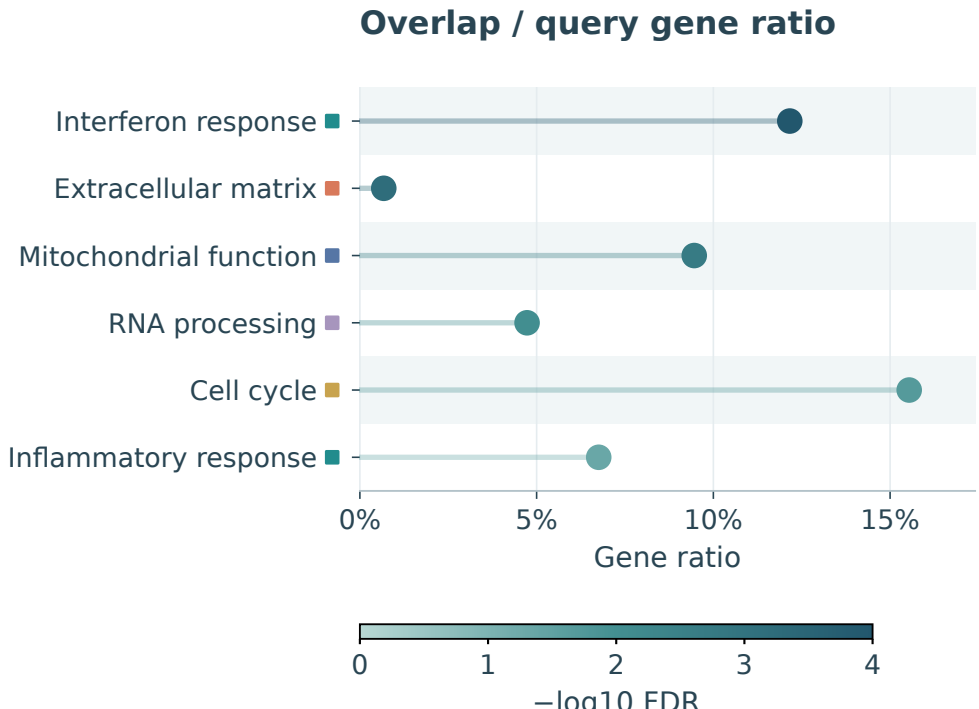
All 1,200 tested genes · stored effects and complete marginal counts

Selected gene expression
Every point is one donor · six per condition



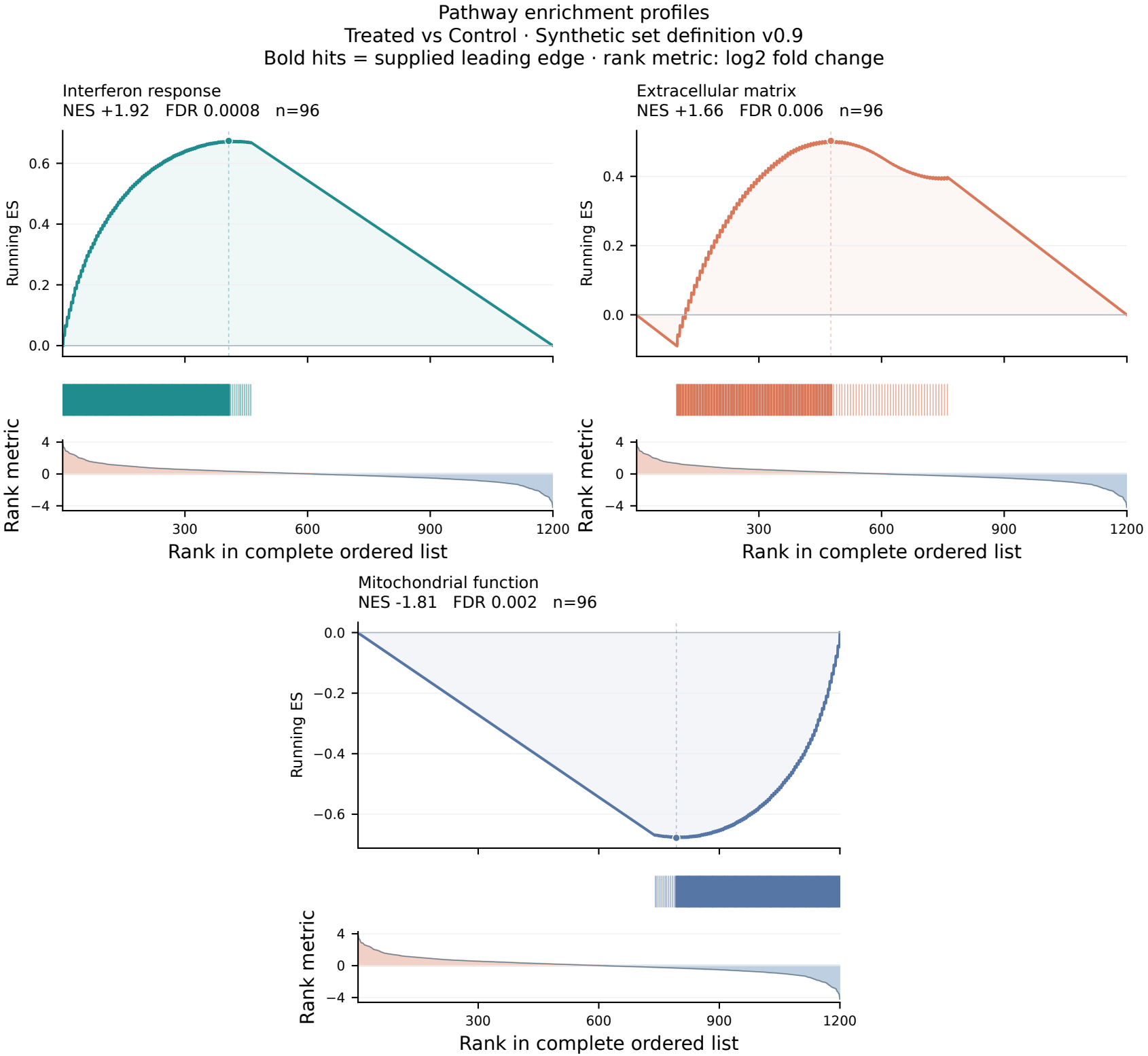
Pathway enrichment

Treated versus Control · 148 query genes / 1200 tested genes



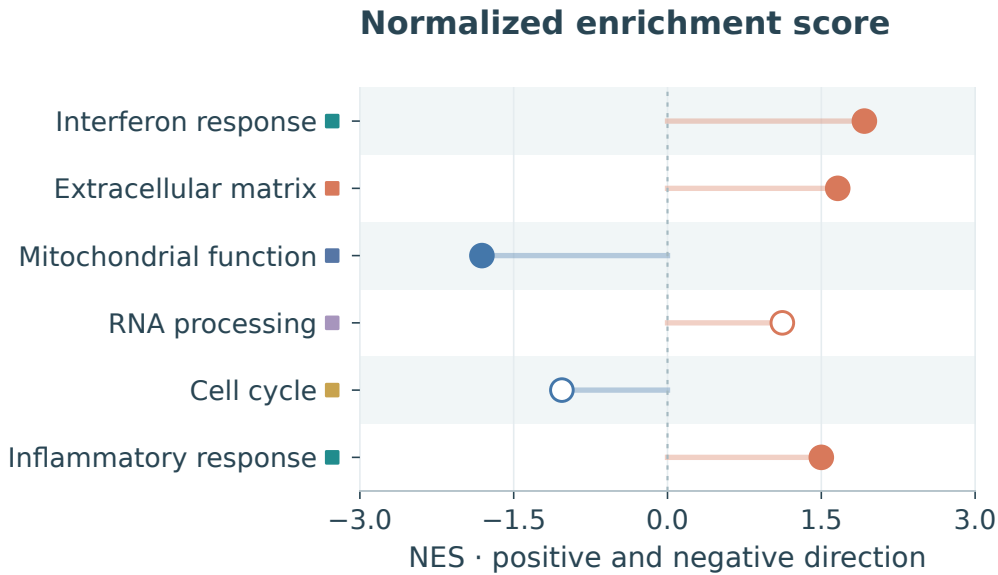
	Genes	FDR	Theme
Interferon response	18	0.0001	Immune
Extracellular matrix	1	0.0006	Matrix
Mitochondrial function	14	0.002	Metabolic
RNA processing	7	0.009	RNA
Cell cycle	23	0.02	Cellular
Inflammatory response	10	0.045	Immune

Filled: FDR ≤ 0.05
Open: FDR > 0.05


Pathway enrichment profiles
Treated vs Control · Synthetic set definition v0.9
Bold hits = supplied leading edge · rank metric: log2 fold change
Interferon response
NES +1.92 FDR 0.0008 n=96
Running ES
Rank metric
Rank in complete ordered list
Extracellular matrix
NES +1.66 FDR 0.006 n=96
Running ES
Rank metric
Rank in complete ordered list
Mitochondrial function
NES -1.81 FDR 0.002 n=96
Running ES
Rank metric
Rank in complete ordered list

Pathway direction

Treated versus Control · ranked genes 1200

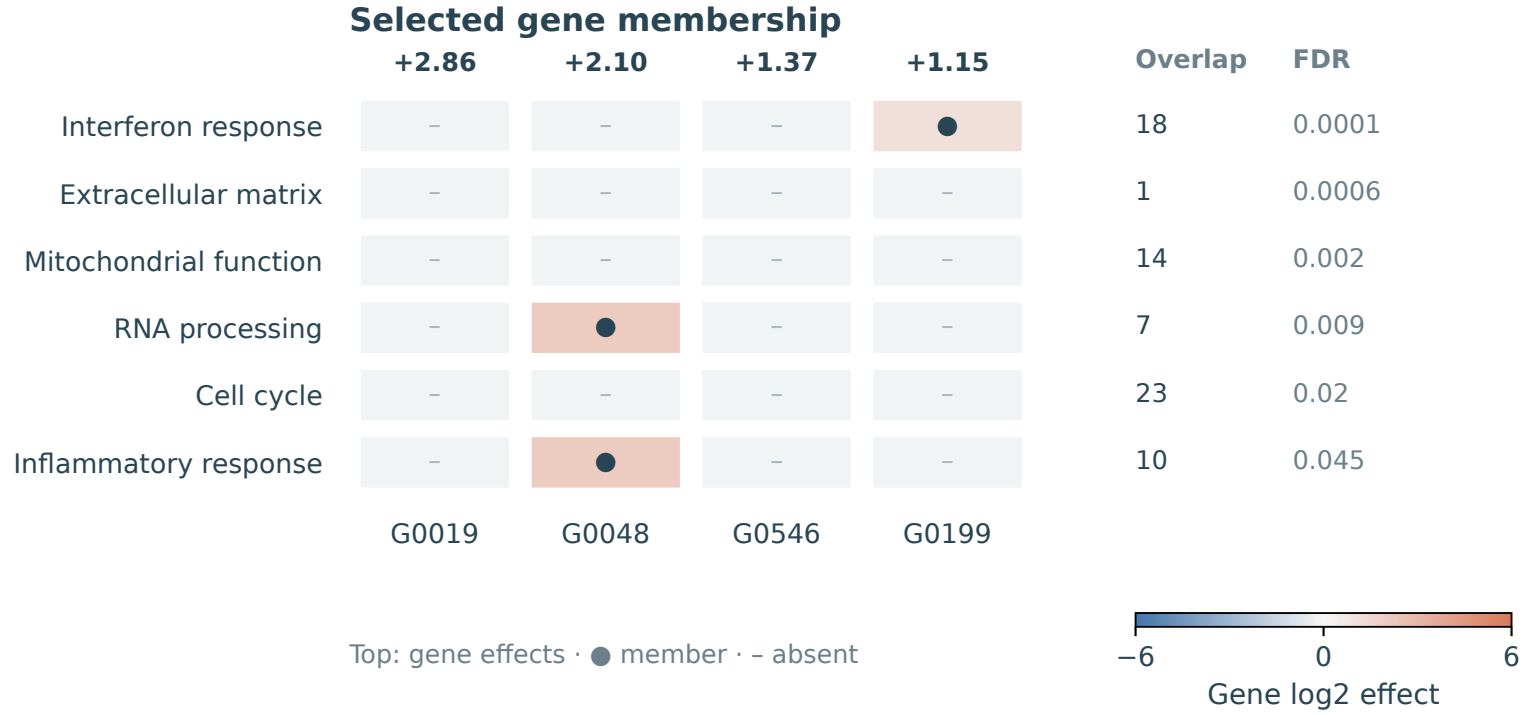


NES	FDR	Theme
+1.92	0.0008	Immune
+1.66	0.006	Matrix
-1.81	0.002	Metabolic
+1.12	0.08	RNA
-1.03	0.12	Cellular
+1.50	0.014	Immune

Filled: FDR ≤ 0.05
Open: FDR > 0.05

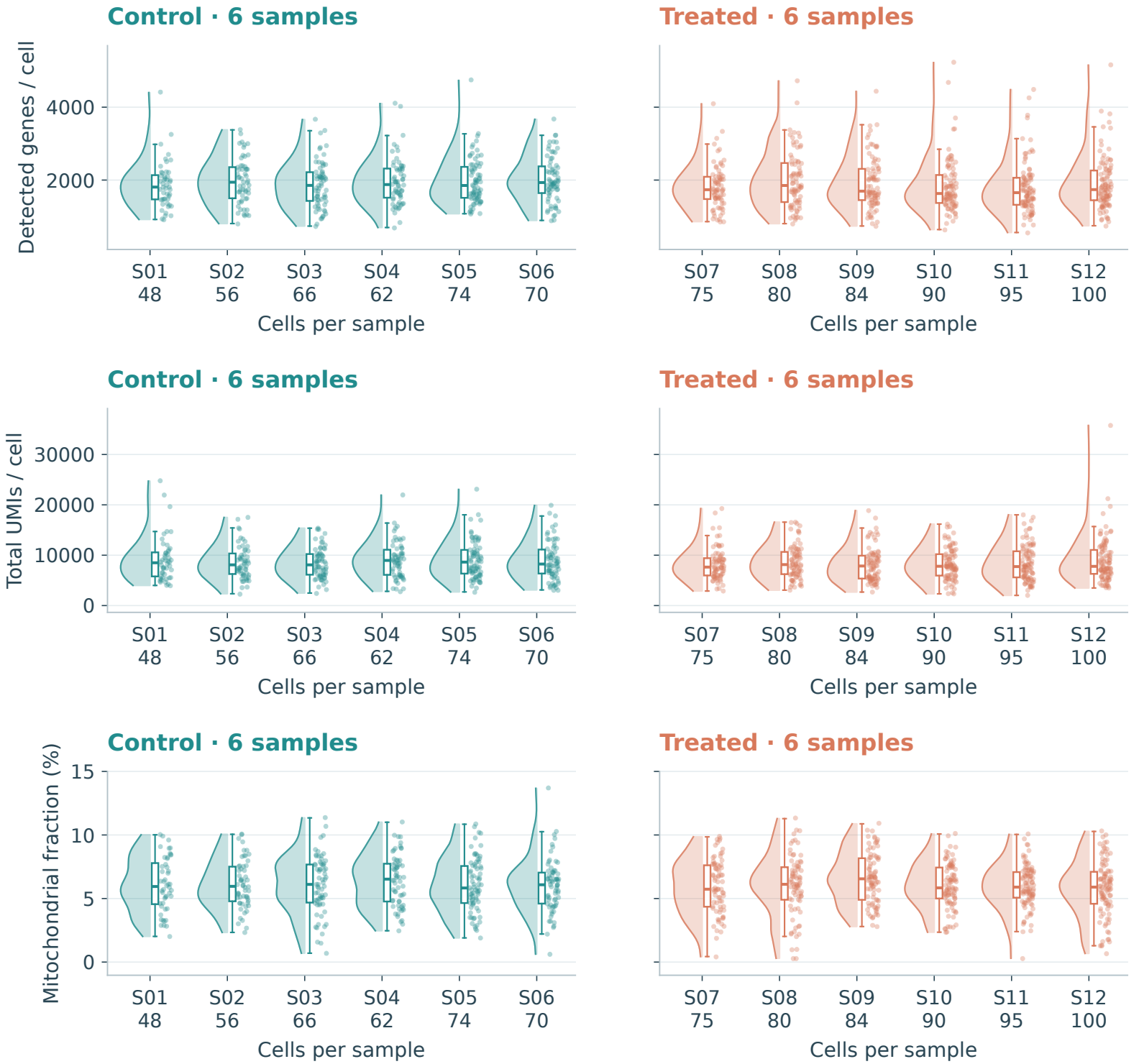
Gene–pathway support

Treated versus Control · 148 query genes / 1200 tested genes



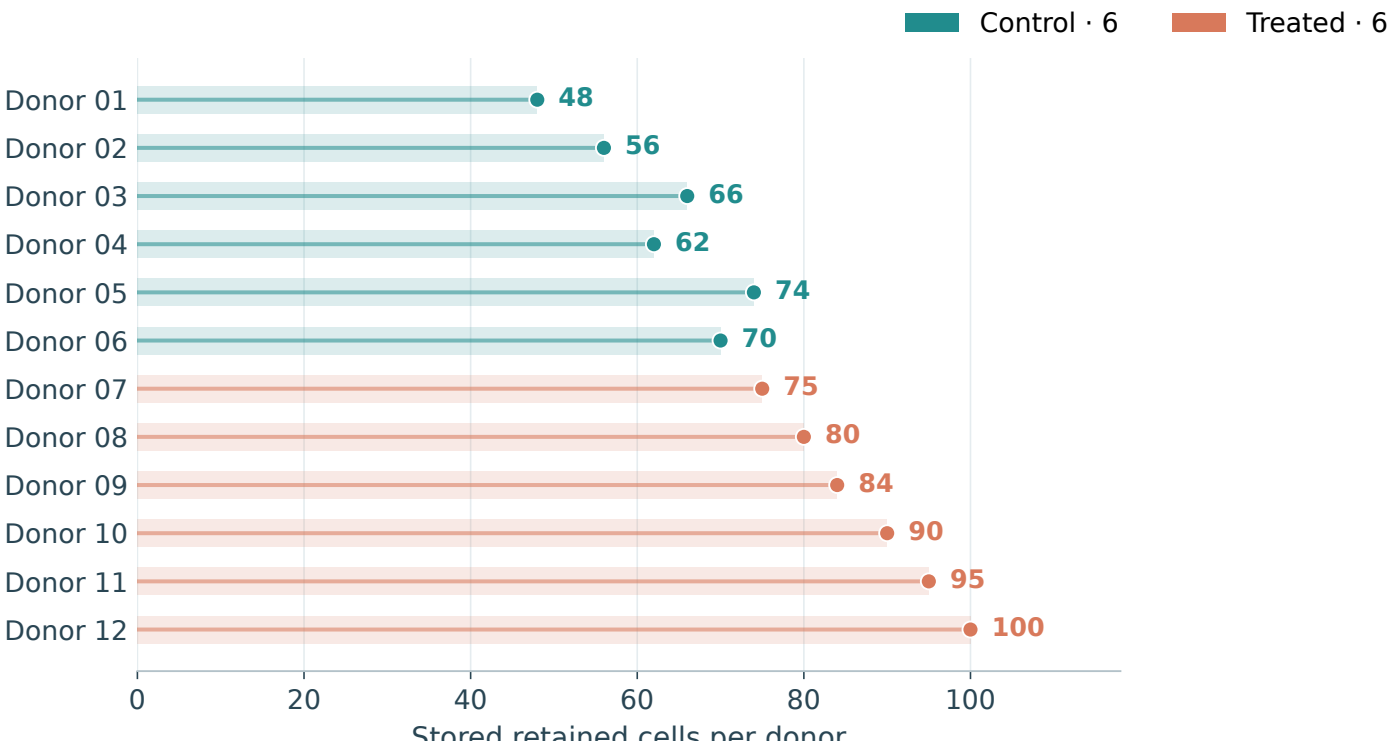
Single-cell quality distributions

12 samples · complete supplied observations · shared metric axes



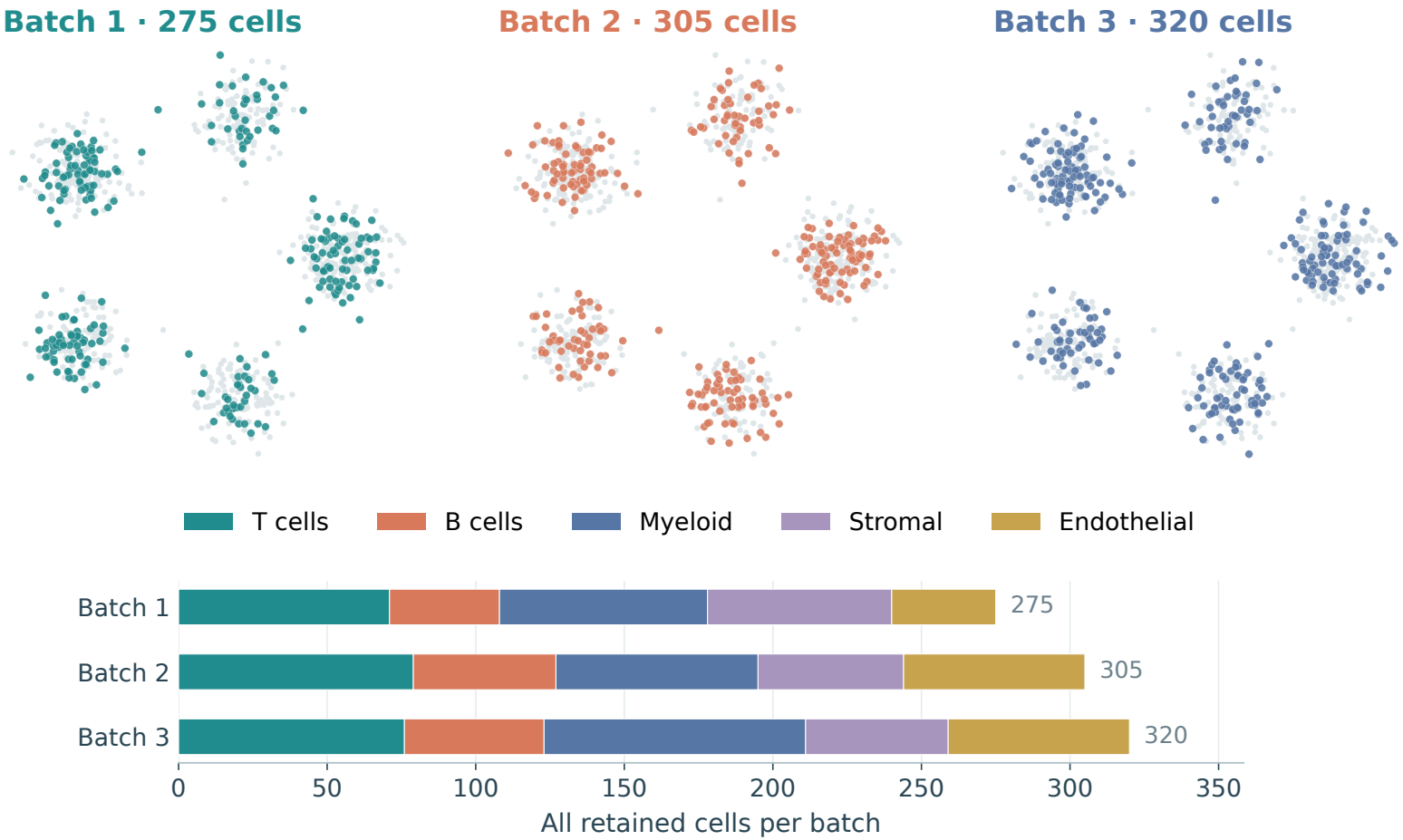
Retained cells by donor

12 supplied units · all counts retained



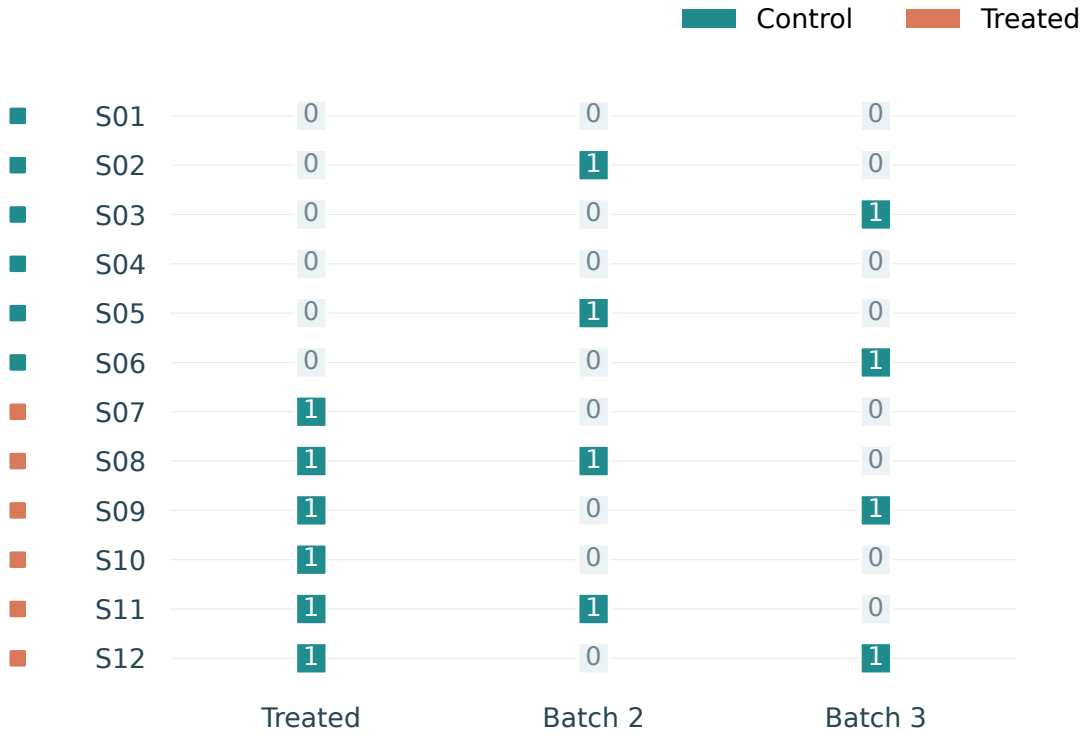
Batch representation across cell identities

Stored coordinates · all cells retained in every background



Donor design and batch coding

12 simulated donors · six per condition

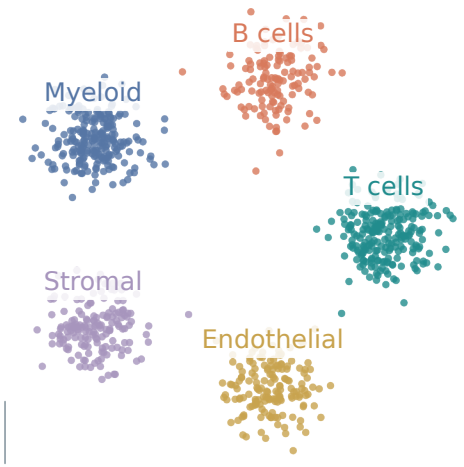


Reference: Control and Batch 1; 0 = absent, 1 = present

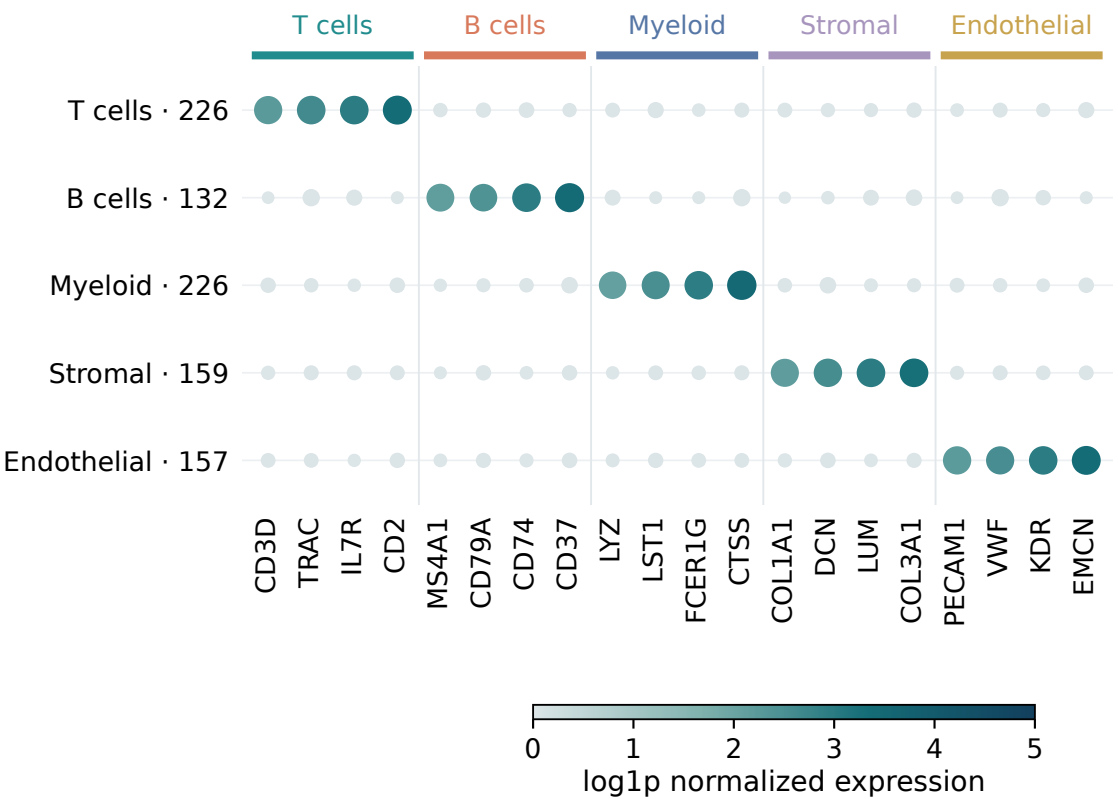
Cell identities and marker evidence

Identity and cross-population marker evidence

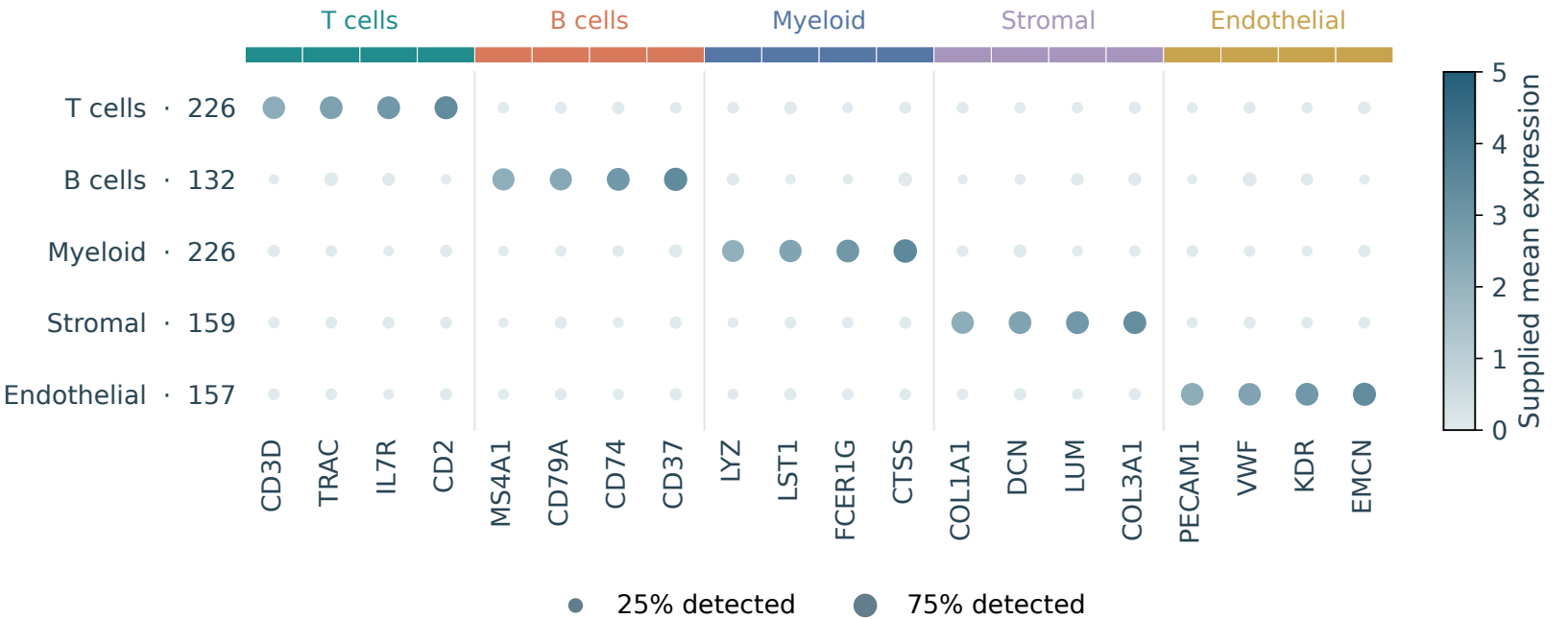
A Cell identities



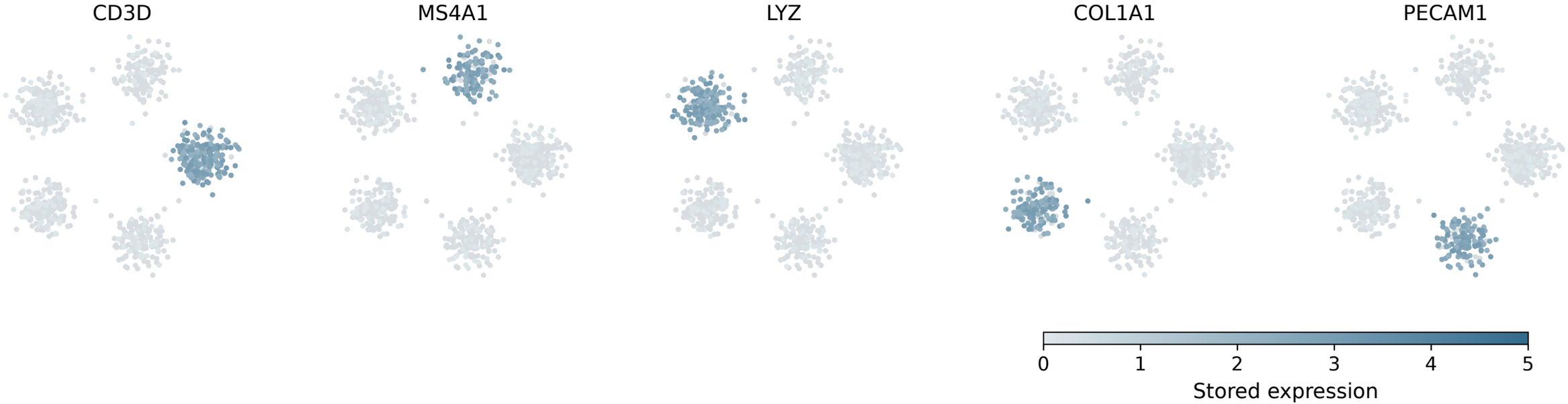
B Marker specificity (mean)



Marker identity and detection



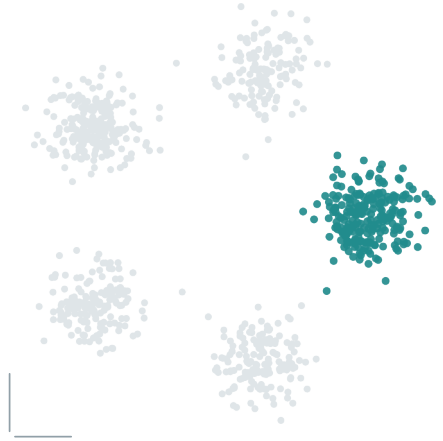
Representative markers · shared coordinates and expression scale



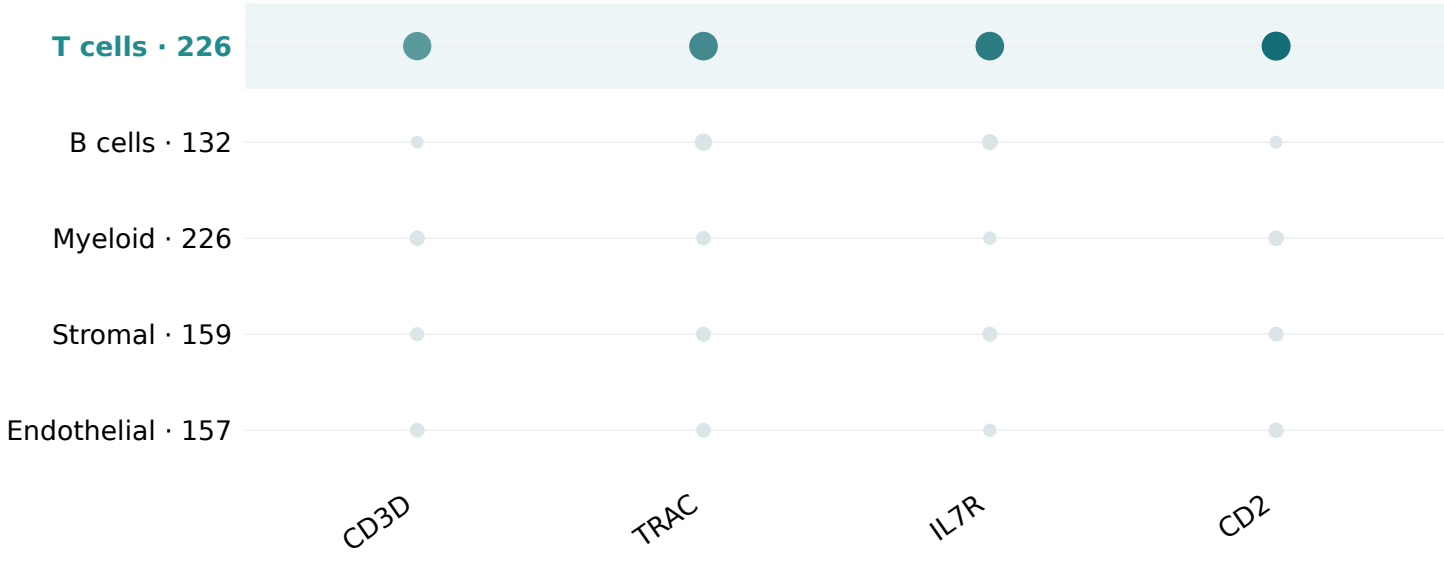
T cells · marker evidence

T cells / 226 of 900 displayed cells / 4 selected markers

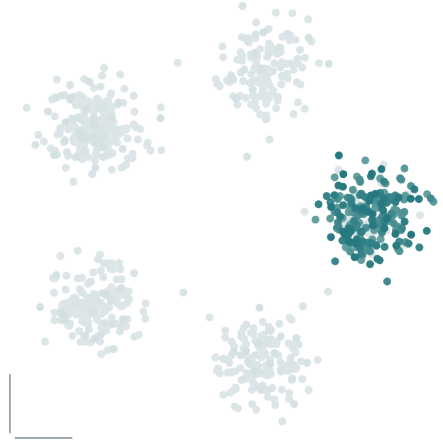
A Population location



B Marker specificity (mean)

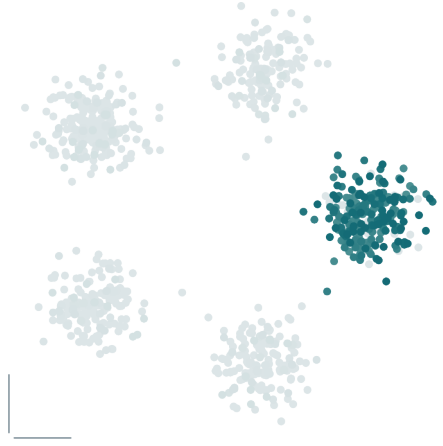


C CD3D



Detected 86% · mean 2.23

D TRAC



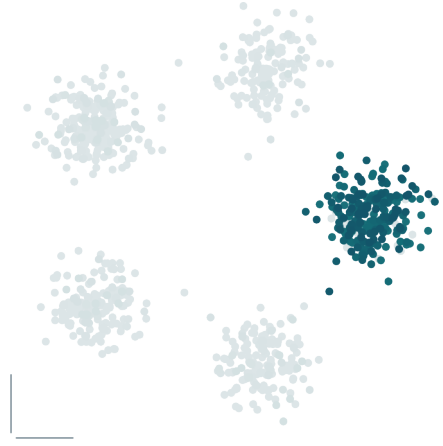
Detected 88% · mean 2.62

E IL7R

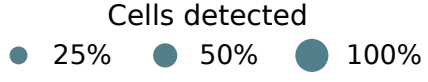


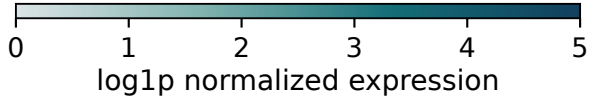
Detected 88% · mean 2.95

F CD2



Detected 91% · mean 3.37

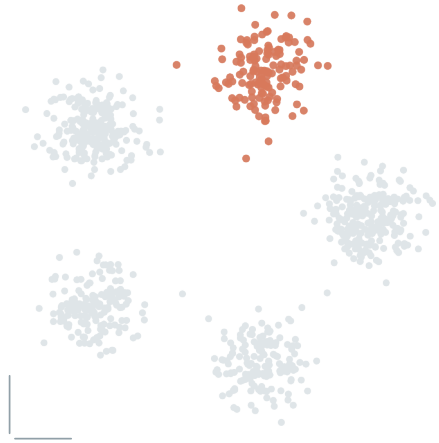




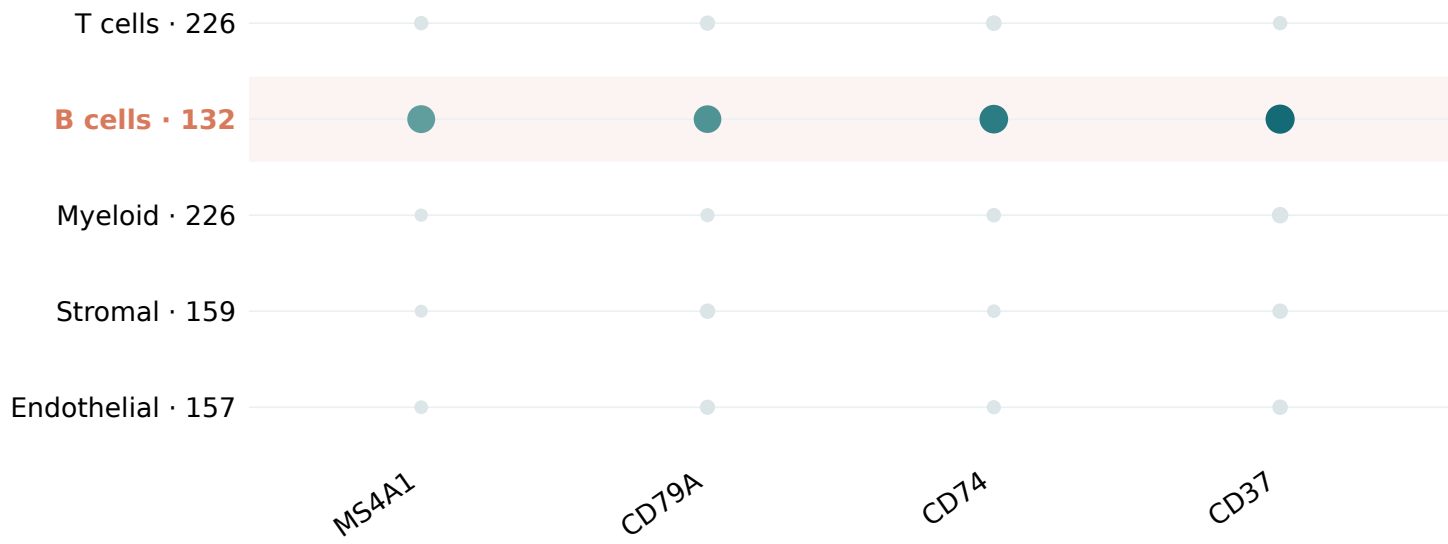
B cells · marker evidence

B cells / 132 of 900 displayed cells / 4 selected markers

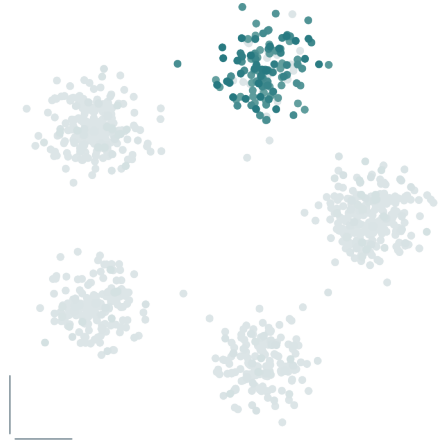
A Population location



B Marker specificity (mean)

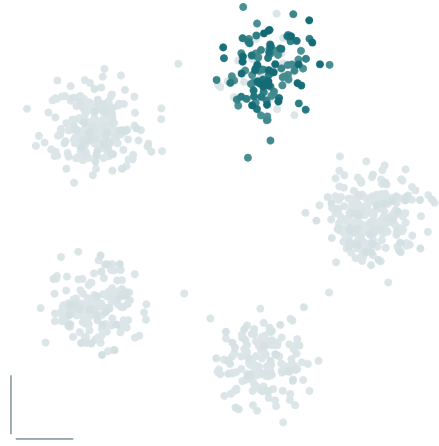


C MS4A1



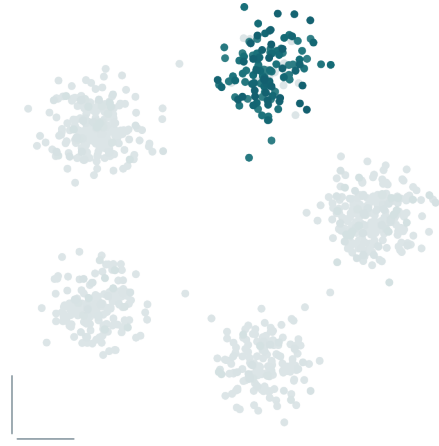
Detected 84% · mean 2.15

D CD79A



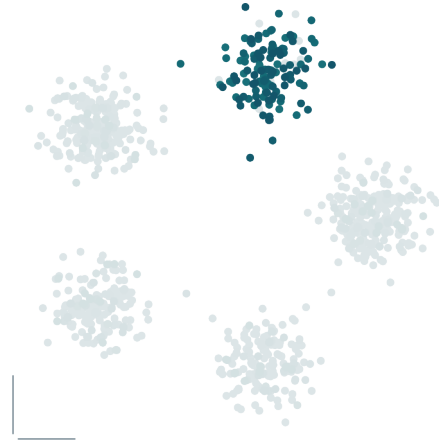
Detected 83% · mean 2.41

E CD74



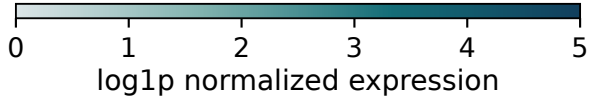
Detected 89% · mean 2.96

F CD37



Detected 92% · mean 3.42





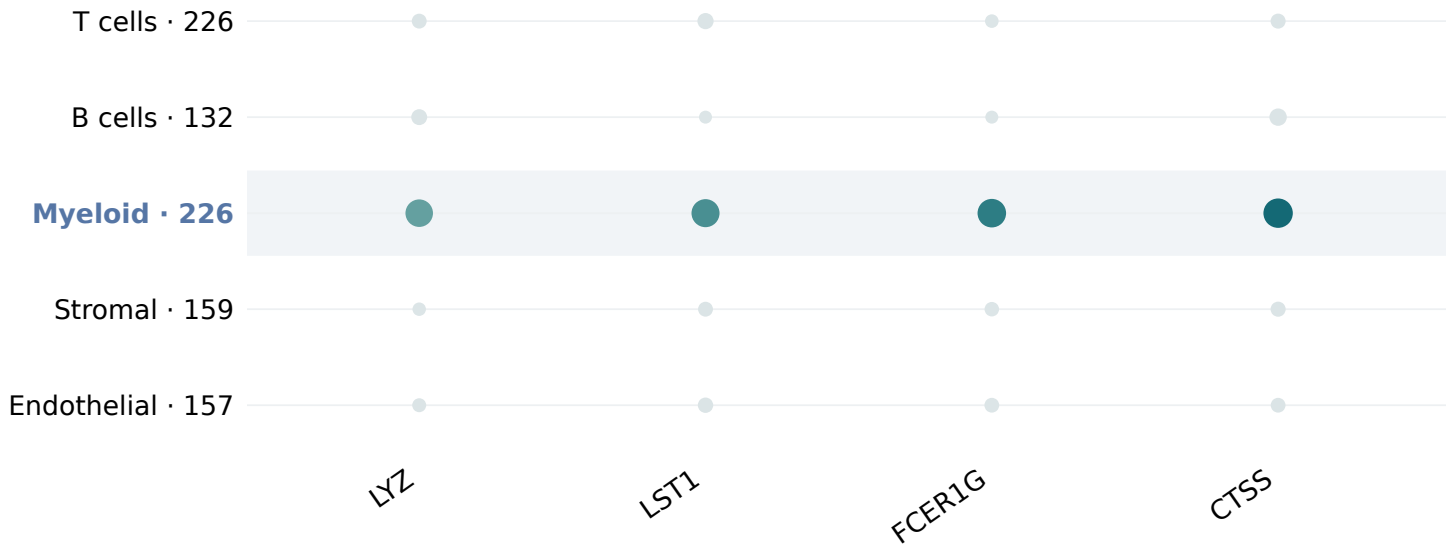
Myeloid · marker evidence

Myeloid / 226 of 900 displayed cells / 4 selected markers

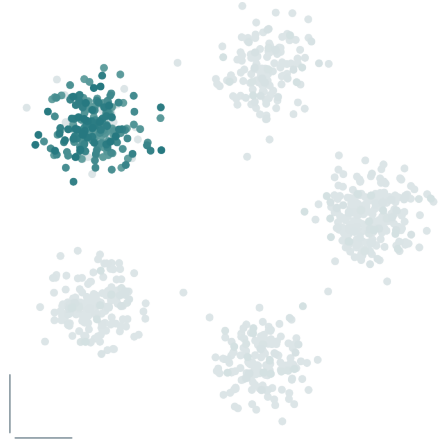
A Population location



B Marker specificity (mean)

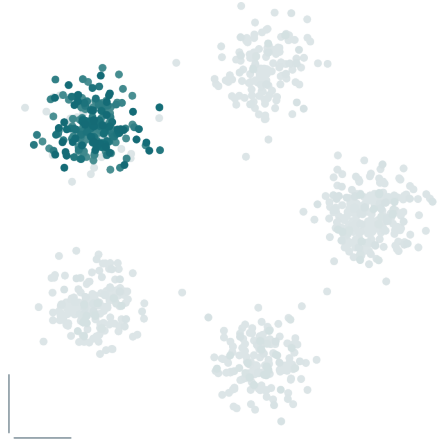


C LYZ



Detected 82% · mean 2.10

D LST1



Detected 85% · mean 2.51

E FCER1G

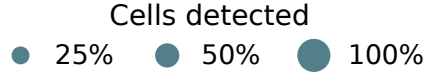


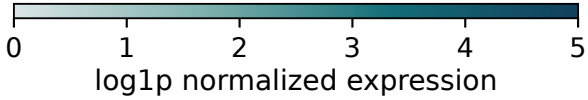
Detected 88% · mean 2.94

F CTSS



Detected 93% · mean 3.48

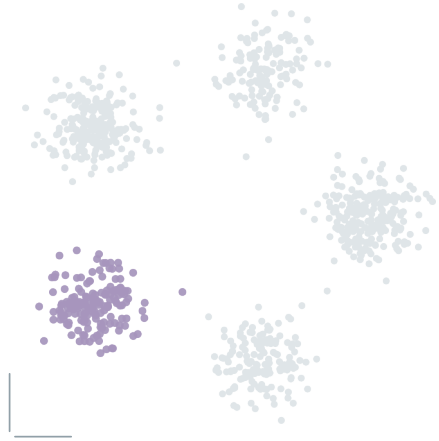




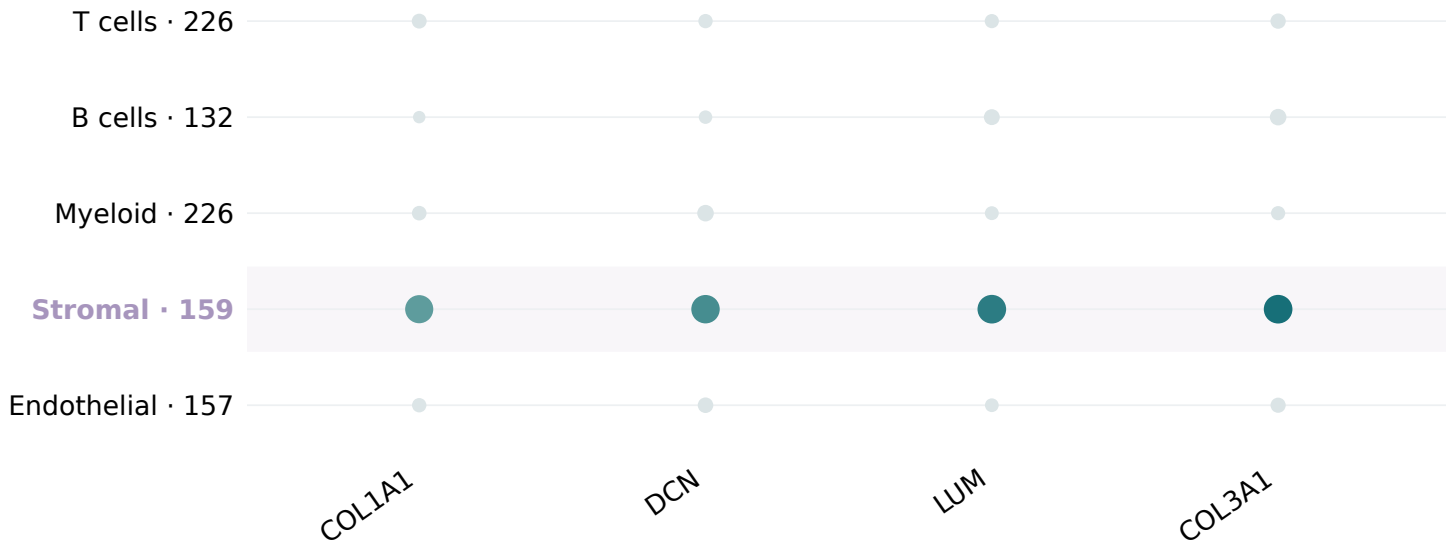
Stromal · marker evidence

Stromal / 159 of 900 displayed cells / 4 selected markers

A Population location



B Marker specificity (mean)

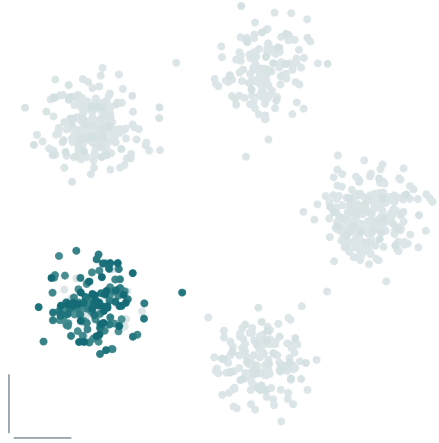


C COL1A1



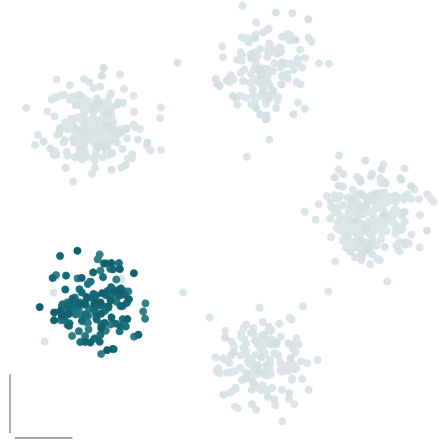
Detected 86% · mean 2.19

D DCN



Detected 87% · mean 2.58

E LUM

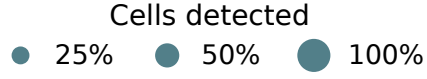


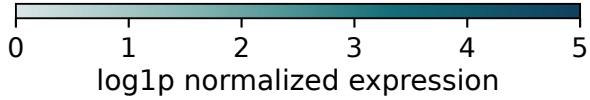
Detected 89% · mean 2.95

F COL3A1



Detected 89% · mean 3.29





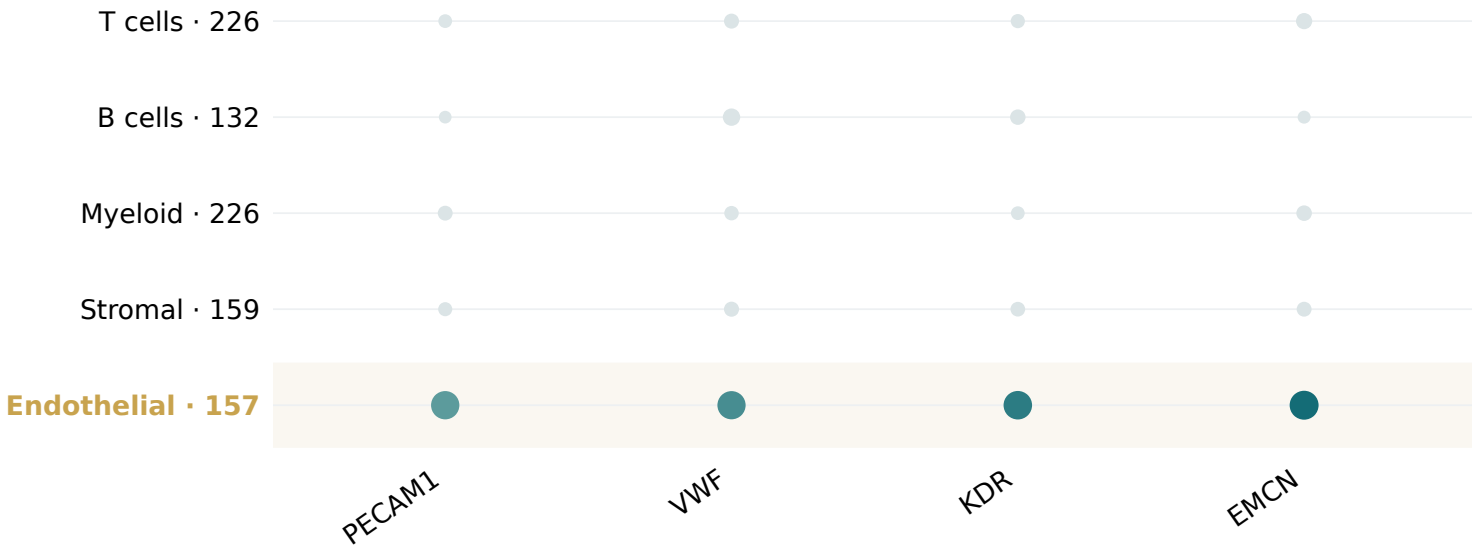
Endothelial · marker evidence

Endothelial / 157 of 900 displayed cells / 4 selected markers

A Population location



B Marker specificity (mean)



C PECAM1



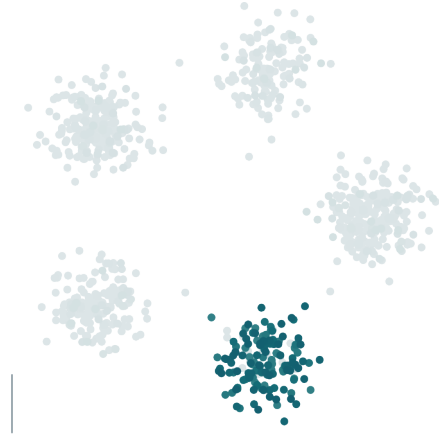
Detected 87% · mean 2.21

D VWF



Detected 86% · mean 2.55

E KDR



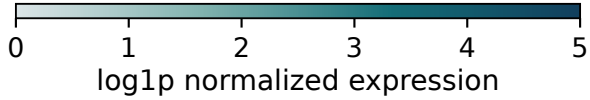
Detected 88% · mean 2.96

F EMCN

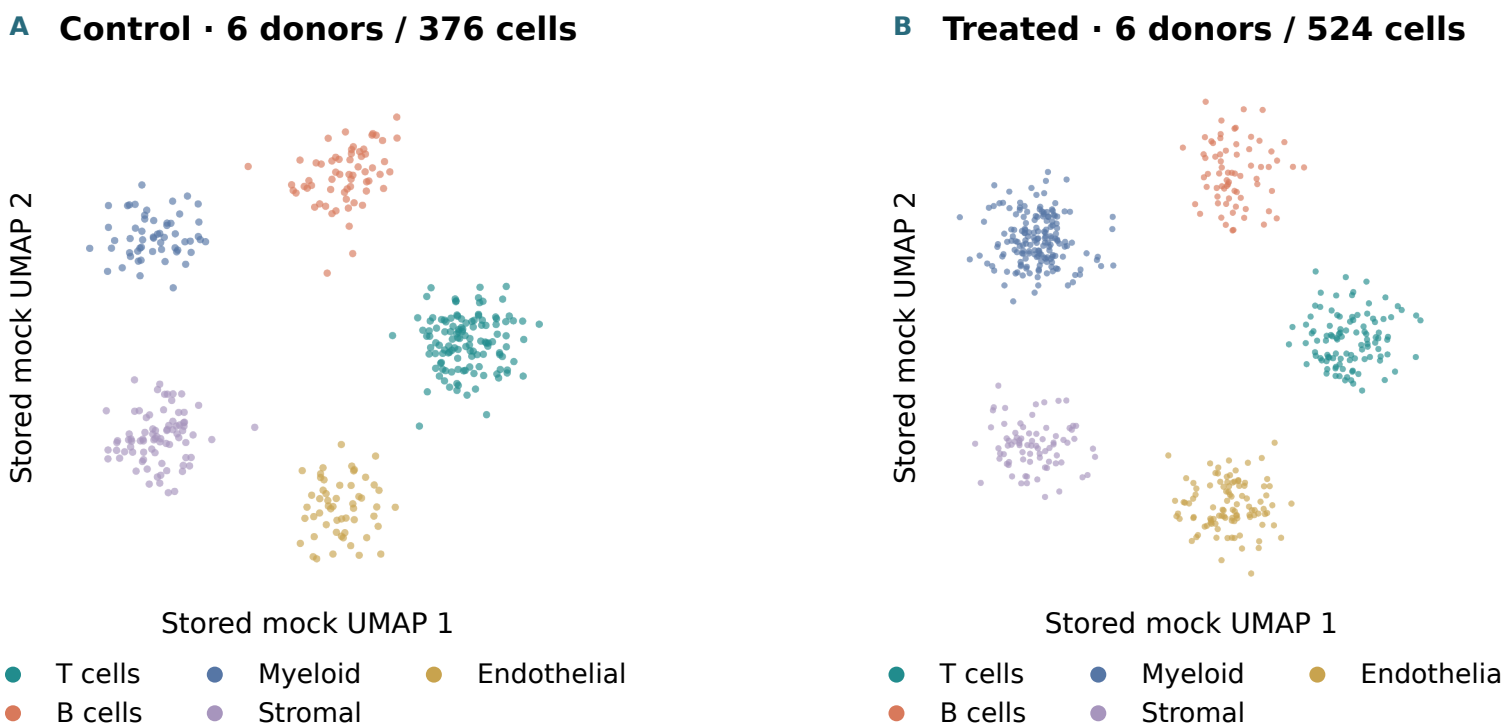


Detected 90% · mean 3.37





Cell identities across conditions
900 cells nested within 12 donors · common coordinates



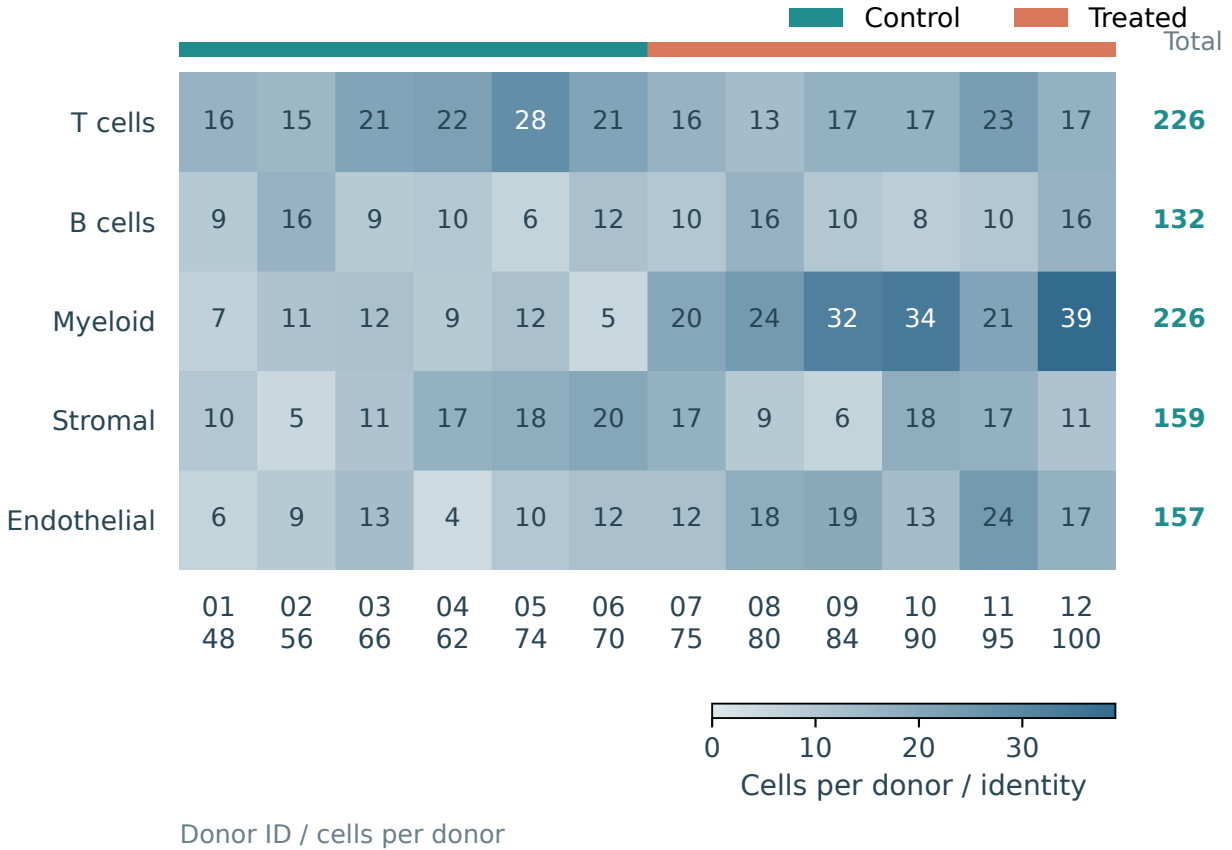
Cell composition across donors

One independent donor per bar and point · denominator: all supplied cells within that donor

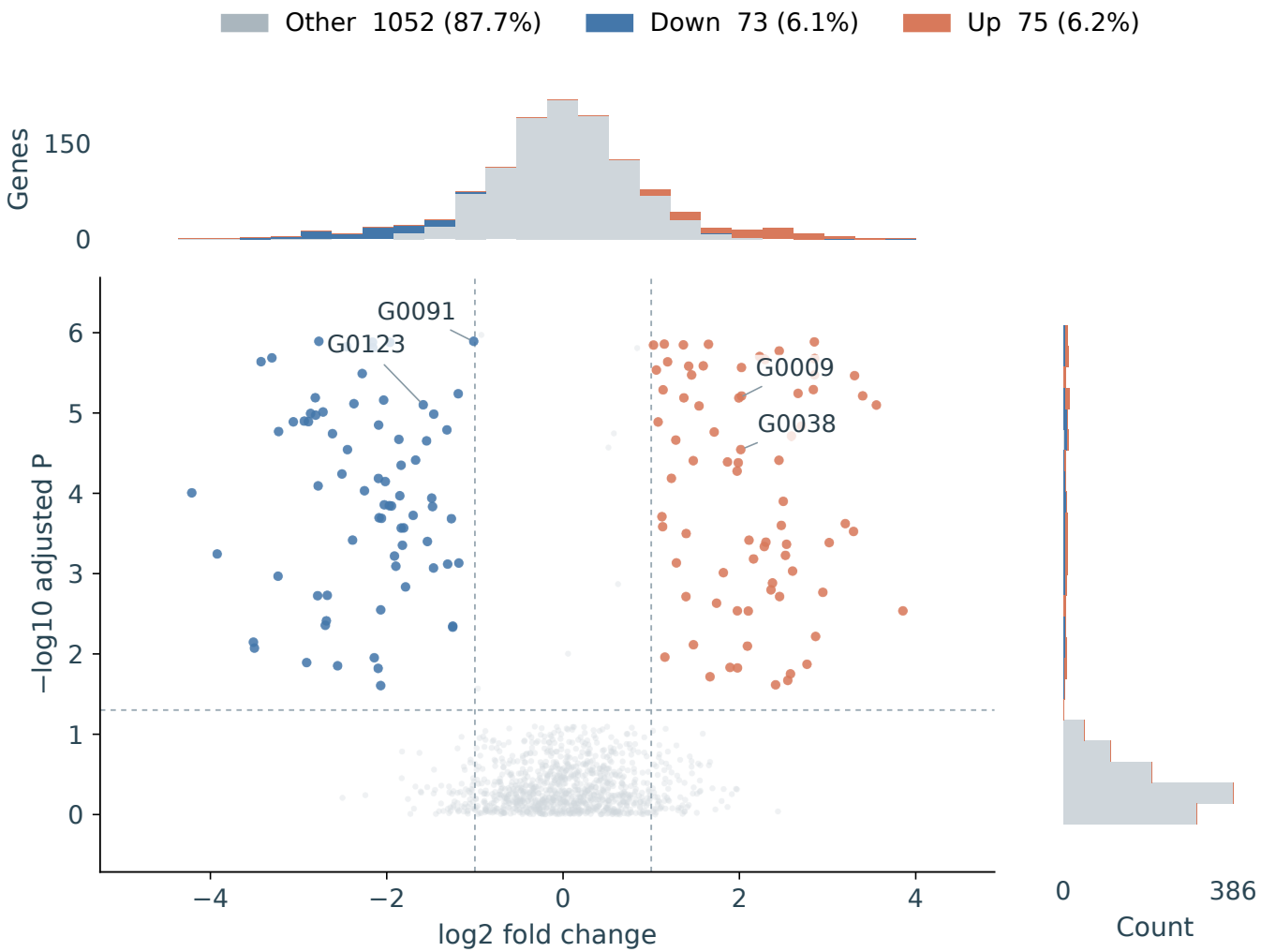


Cell support across independent donors

12 simulated donors · six per condition



Myeloid · differential expression

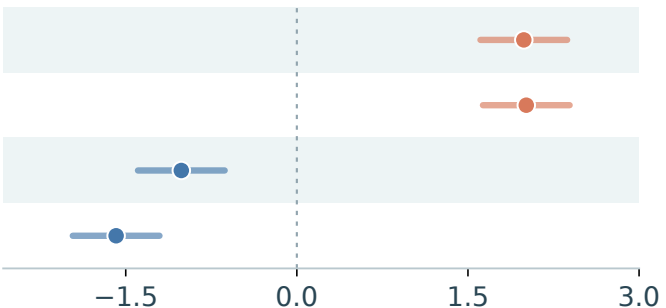


All 1,200 tested genes · stacked counts · FDR ≤ 0.05 and |log2FC| ≥ 1

Myeloid donor effects

Treated versus Control / Myeloid donor-level

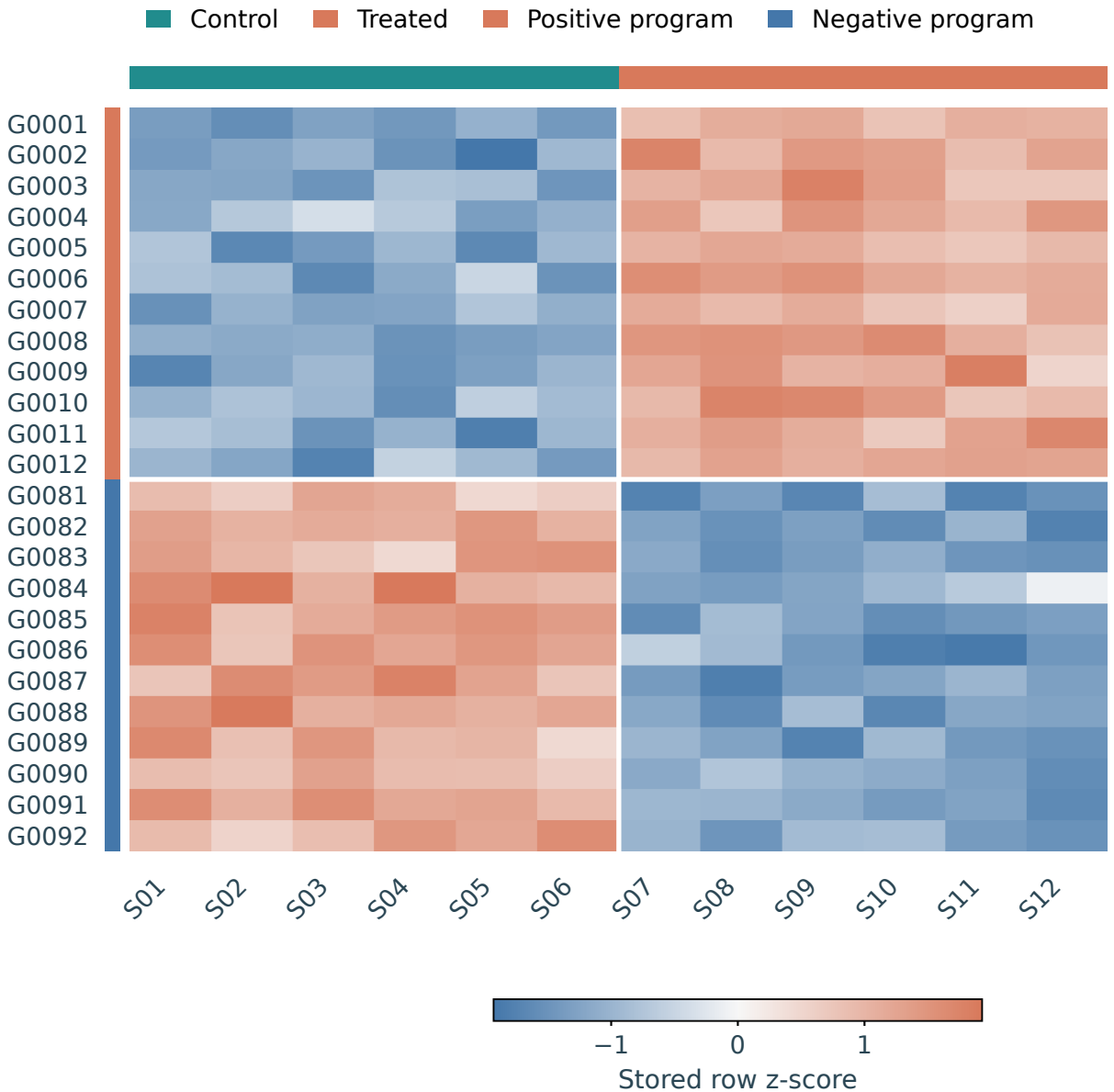
Gene	log2 fold change	Estimate [95% interval]
G0009		+1.99 [+1.61, +2.37]
G0038		+2.01 [+1.63, +2.39]
G0091		-1.01 [-1.39, -0.63]
G0123		-1.58 [-1.96, -1.20]



Points: supplied estimates · Lines: supplied 95% intervals

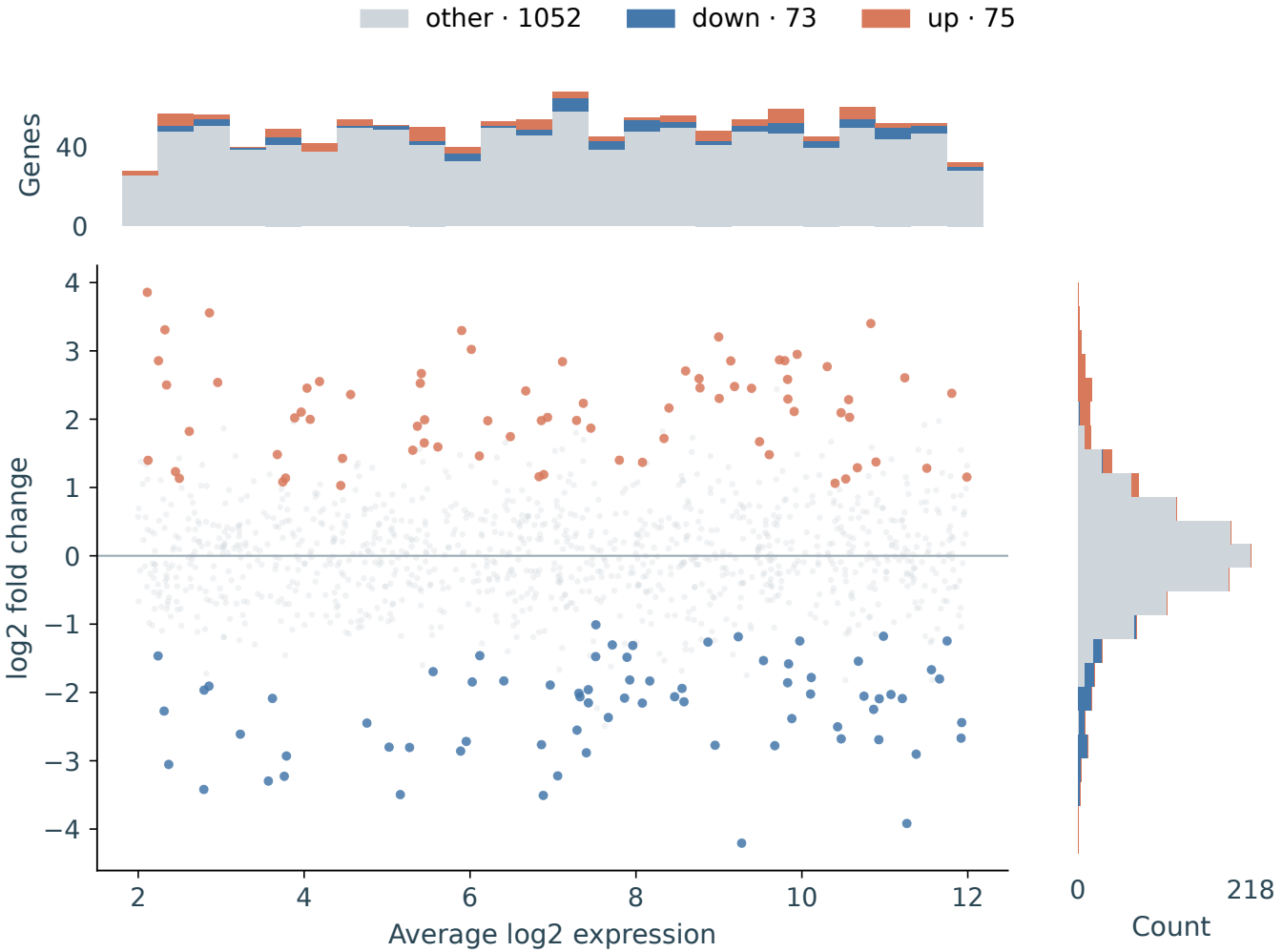
Myeloid · selected expression programs

24 selected features · 12 samples



Software test fixture; not biological evidence

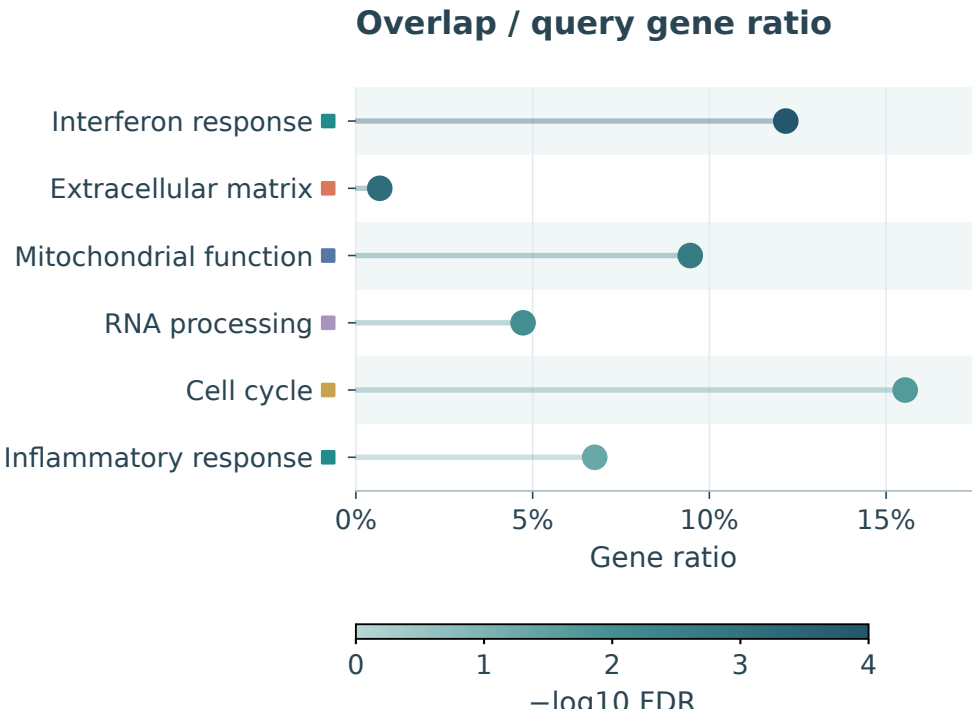
Myeloid · expression level and effect size



All 1,200 tested genes · stored effects and complete marginal counts

Myeloid · Pathway enrichment

Treated versus Control · 148 query genes / 1200 tested genes

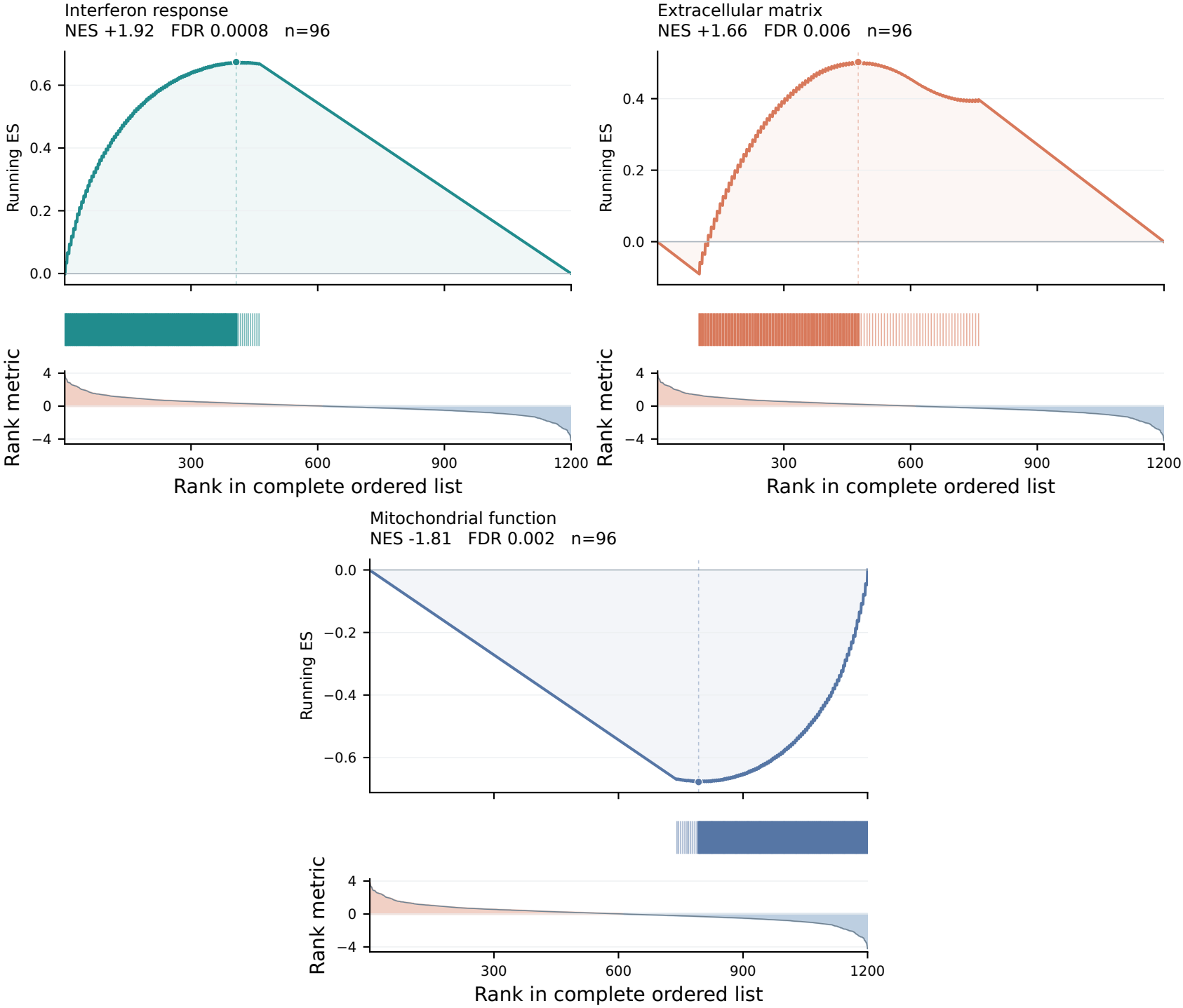


	Genes	FDR	Theme
Interferon response	18	0.0001	Immune
Extracellular matrix	1	0.0006	Matrix
Mitochondrial function	14	0.002	Metabolic
RNA processing	7	0.009	RNA
Cell cycle	23	0.02	Cellular
Inflammatory response	10	0.045	Immune

Filled: FDR ≤ 0.05
Open: FDR > 0.05

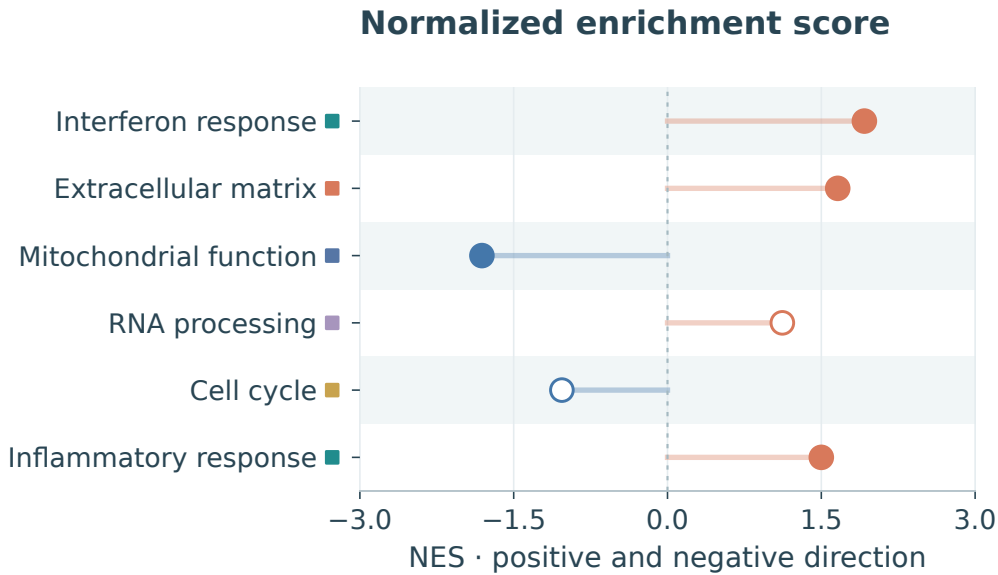
Software test fixture; not biological evidence

Myeloid · Pathway enrichment profiles
Treated vs Control · Synthetic set definition v0.9
Bold hits = supplied leading edge · rank metric: log2 fold change



Myeloid · Pathway direction

Treated versus Control · ranked genes 1200



	NES	FDR	Theme
Interferon response	+1.92	0.0008	Immune
Extracellular matrix	+1.66	0.006	Matrix
Mitochondrial function	-1.81	0.002	Metabolic
RNA processing	+1.12	0.08	RNA
Cell cycle	-1.03	0.12	Cellular
Inflammatory response	+1.50	0.014	Immune

Filled: FDR ≤ 0.05
Open: FDR > 0.05

Myeloid · Gene–pathway support

Treated versus Control · 148 query genes / 1200 tested genes

Selected gene membership

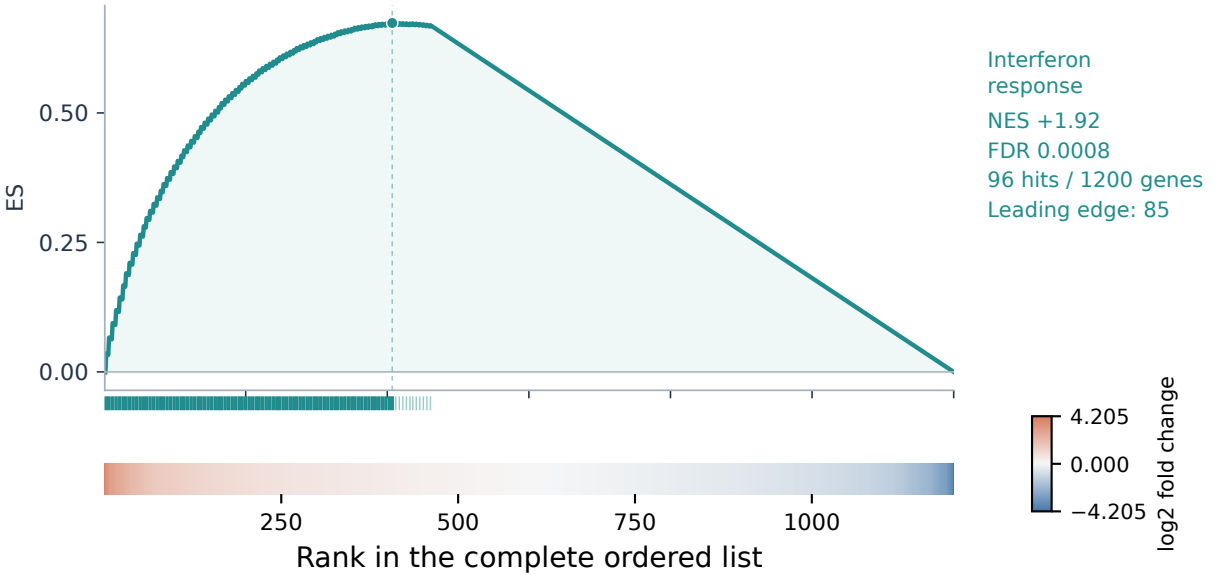
	+2.86	+2.10	+1.37	+1.15	Overlap	FDR
Interferon response	–	–	–	●	18	0.0001
Extracellular matrix	–	–	–	–	1	0.0006
Mitochondrial function	–	–	–	–	14	0.002
RNA processing	–	●	–	–	7	0.009
Cell cycle	–	–	–	–	23	0.02
Inflammatory response	–	●	–	–	10	0.045
	G0019	G0048	G0546	G0199		

Top: gene effects · ● member · – absent



Interferon response

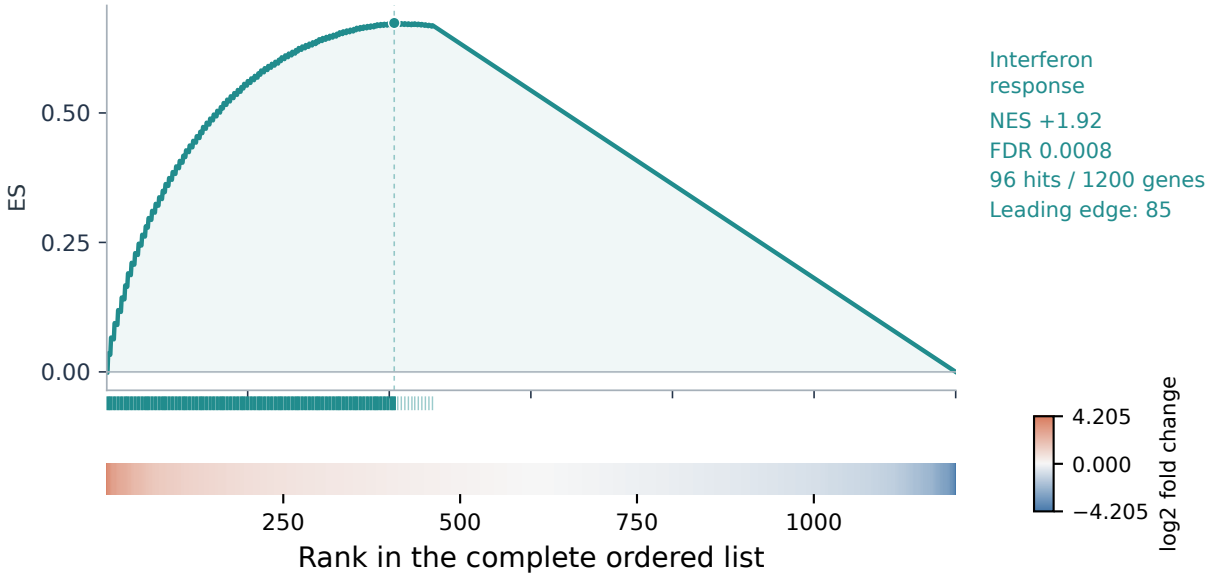
Treated vs Control / exact supplied enrichment tracks



Hit rugs retain exact ranks; bold hits are supplied leading edge. Synthetic set definition v0.9. No ES, NES, FDR or membership recomputation.

Myeloid · Interferon response

Treated vs Control / exact supplied enrichment tracks



Hit rugs retain exact ranks; bold hits are supplied leading edge. Synthetic set definition v0.9. No ES, NES, FDR or membership recomputation.

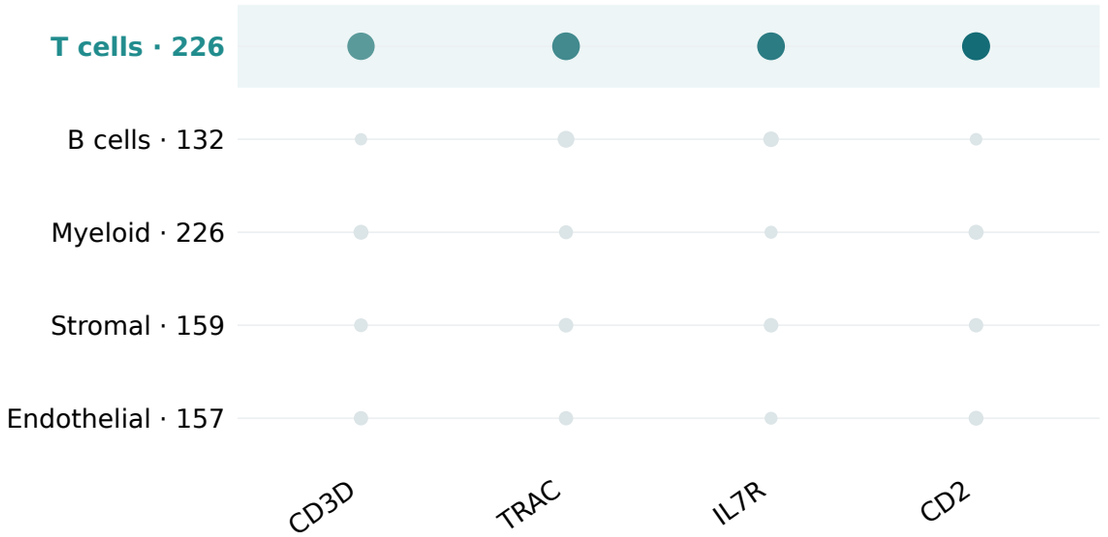
T cells · marker evidence

T cells / 226 of 900 displayed cells / 4 selected markers

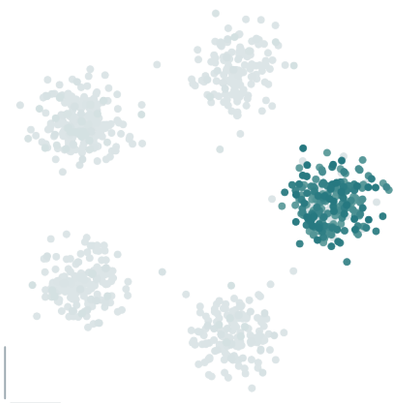
A Population location



B Marker specificity (mean)

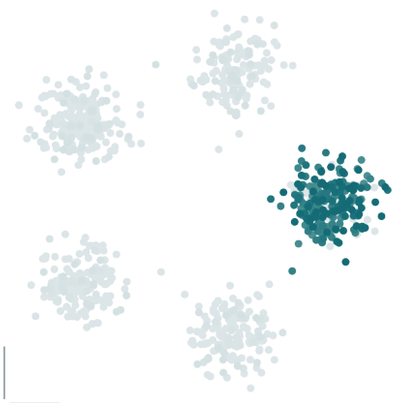


C CD3D



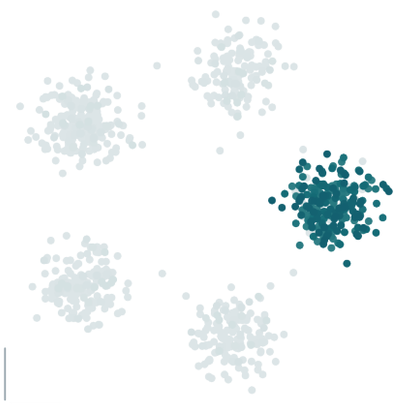
Detected 86% · mean 2.23

D TRAC



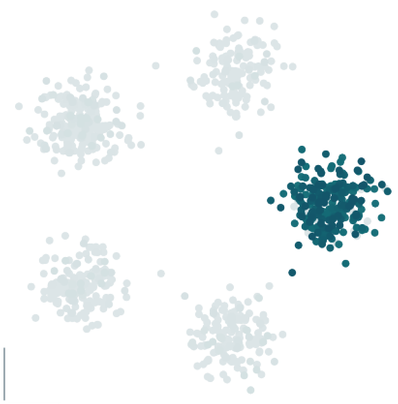
Detected 88% · mean 2.62

E IL7R

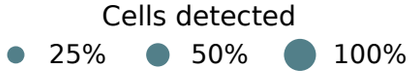


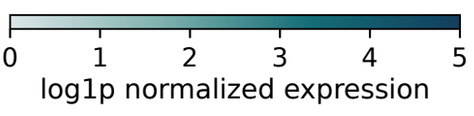
Detected 88% · mean 2.95

F CD2



Detected 91% · mean 3.37

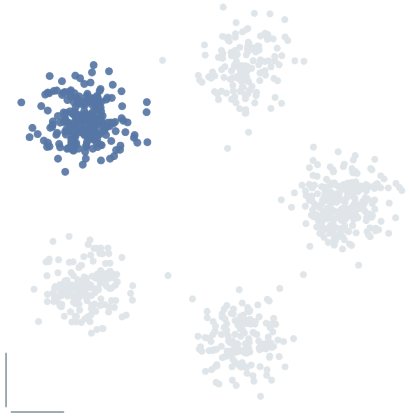




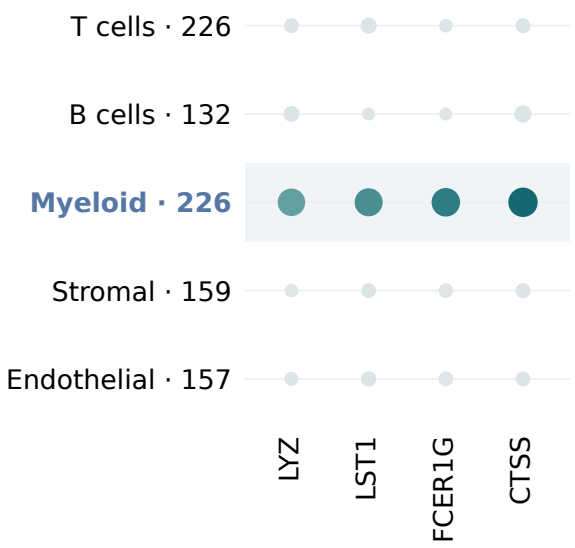
Myeloid · marker evidence

Myeloid / 226 of 900 displayed cells / 4 selected markers

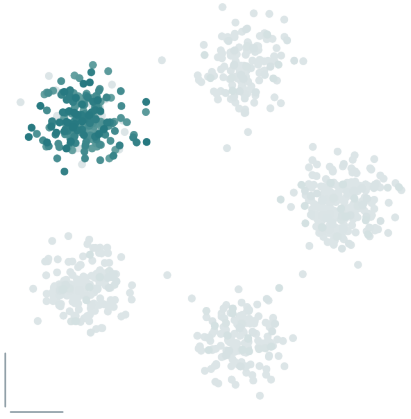
A Population location



B Specificity

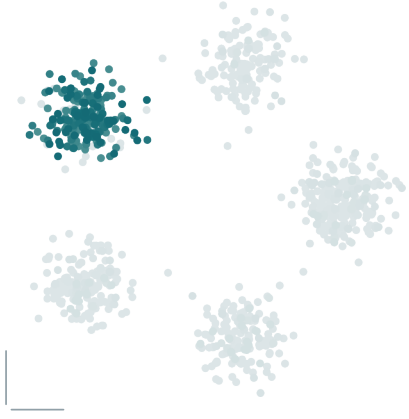


C LYZ



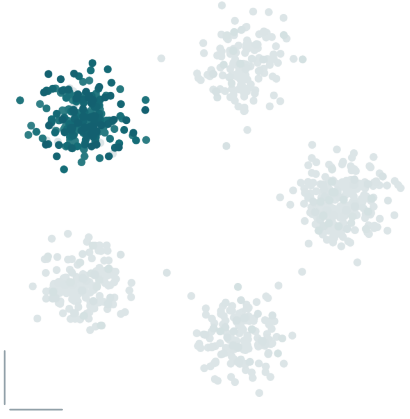
Detected 82% · mean 2.10

D LST1



Detected 85% · mean 2.51

E FCER1G

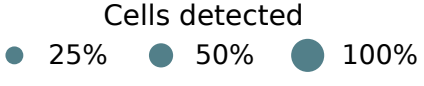


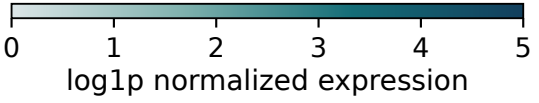
Detected 88% · mean 2.94

F CTSS



Detected 93% · mean 3.48





Pathway enrichment profiles
Treated vs Control · Synthetic set definition v0.9
Bold hits = supplied leading edge · rank metric: log2 fold change

